Hybrid Hidden Markov Model for Modeling Equity Excess Growth Rate Dynamics: A Discrete-State Approach with Jump-Diffusion

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Abstract

Synthetic equity return paths underpin stress testing, portfolio validation, and training data for machine-learning pipelines, yet parametric, neural, and semi-Markov generators rarely reproduce the single-asset stylized facts of heavy tails, regime persistence, and volatility clustering in a single generator. We developed a hybrid hidden Markov framework that discretized excess growth rates into states defined by quantiles of a fitted Laplace distribution, used a location-scale Student-t emission within each state, and augmented regime switching with a Poisson jump-duration mechanism that enforced realistic tail-state dwell times. We evaluated the framework on a decade of SPY history with a one-year hold-out, benchmarking against bootstrap, Gaussian, Laplace, GARCH(1,1), a Gated Recurrent Unit, a hidden semi-Markov model, and a no-jump variant of the framework on distributional, tail, and temporal metrics. The hybrid marginals reproduced the stylized facts in-sample and out-of-sample, and the jump mechanism was the only addition that reduced absolute-growth autocorrelation meaningfully below the i.i.d. floor without sacrificing distributional fit. To extend the framework to multi-asset workflows, we composed per-asset paths under a single-index model using a closed-form variance correction that rescaled the per-asset generator draw before insertion into the market factor; the correction recovered the calibrated factor loading on a 424-asset US equity and exchange-traded-fund universe while preserving heavy-tailed character at tail-risk accuracy comparable to dedicated residual fits. We also surfaced a daily-rebalancing artifact in which iid residuals inflate the diversification return, and corrected it with a uniform location shift anchored to a long-run prior. All code and per-asset fitted marginals were released alongside the paper.

Keywords: hidden Markov model; jump-diffusion; single-index model; synthetic financial data; variance correction; Value-at-Risk backtest; volatility clustering; heavy-tailed marginals.

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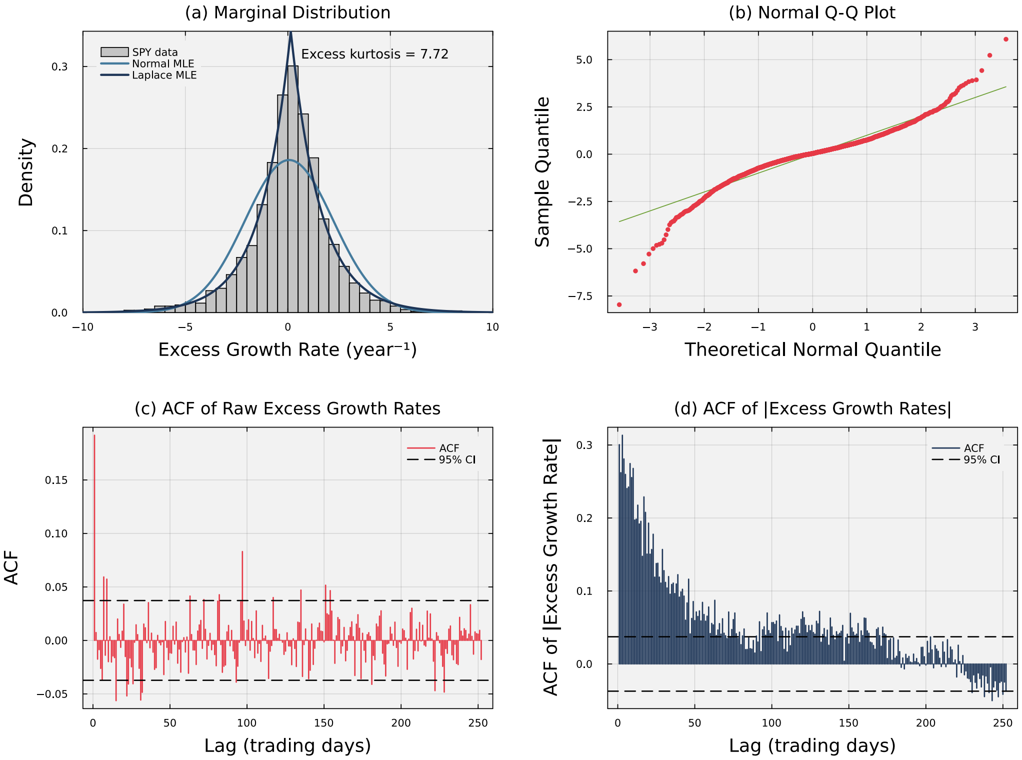

Figure 1: **Empirical stylized facts of SPY daily excess growth rates** (2014–2024)**.** Panel (a) shows the leptokurtic marginal density; the Laplace maximum-likelihood fit tracks the empirical peak and tails while the Gaussian fit does not. Panel (b) confirms heavy tails through a normal quantile-quantile plot, with departure from the diagonal at both extremes. Panel (c) shows the autocorrelation function of raw excess growth rates, which sits near zero across all lags and is consistent with the efficient markets hypothesis. Panel (d) shows the autocorrelation function of absolute excess growth rates, which decays slowly and signals the persistent volatility clustering that the jump-duration extension is designed to reproduce.

1 Introduction

Generating synthetic financial time series that faithfully preserve the statistical properties of real market data is a central challenge in quantitative finance. Synthetic paths are used for stress testing risk models against plausible but unobserved market scenarios, validating portfolio optimization algorithms, augmenting limited training sets for machine-learning pipelines, and Monte Carlo backtesting of allocators across many independent draws [1, 2]. Empirical equity excess growth rates exhibit three statistical regularities a faithful generative framework must reproduce together: a heavy-tailed (leptokurtic) marginal distribution, negligible linear autocorrelation of raw returns, and persistent volatility clustering, in which large changes are followed by large changes of either sign and small changes by small changes [3–5]. All three are visible in SPY daily excess growth rates over 2014–2024 (Fig. 1): the marginal density is sharply peaked with heavy tails that a Gaussian fit cannot reach, the quantile-quantile plot bends away from the Gaussian line at both ends, and the autocorrelation of absolute growth rates decays slowly while the autocorrelation of raw growth rates sits near zero. These regularities, called *stylized facts*, are the benchmark for synthetic financial data.

Existing generative approaches each excel at one of these regularities but fail at others. The GARCH conditional-variance family [6, 7] captures volatility clustering but leaves light-tailed marginals unless fed a fat-tailed innovation and represents neither discrete regimes nor sudden jumps [8]. Stochastic-volatility and jump-diffusion models [9–11] enrich tail behavior but offer no mechanism for persistence after a jump. Deep generative models [12–14] can learn complex marginals yet routinely miss the absolute-return autocorrelation that defines volatility clustering. Hidden Markov models [15] offer a probabilistic regime-switching structure, yet standard HMMs revert too quickly to central regimes after extreme events and fail to dwell in high-volatility tail states long enough to reproduce the empirical clustering profile [16, 17]. Bridging this gap requires a framework that preserves distributional fidelity and temporal structure together while remaining computationally scalable to large asset universes.

In this study, we developed a hybrid hidden Markov framework that discretized continuous excess growth rates into Laplace quantile-defined states and augmented regime switching with a two-parameter Poisson jump-duration mechanism. The Laplace partition lets the transition matrix be estimated by direct frequentist counting, bypassing the iterative Baum-Welch algorithm [18, 19] and its sensitivity to initialization, and scales to a 424-asset pipeline. The Poisson override forces the chain to dwell in high-volatility tail states for empirically realistic durations, closing the volatility-clustering gap that standard HMMs leave open. Within each state we used a Student-t emission with $\nu = 5$ degrees of freedom, which matched the empirical kurtosis to within 1.3% on SPY where Normal emissions underestimated by approximately 29% (Table S3). We validated the framework on a 424-ticker US equity and ETF universe calibrated against the 2014–2024 SPY history, with 2025 held out for out-of-sample evaluation, scoring synthetic paths against seven alternative generators (bootstrap, Gaussian, Laplace, GARCH(1,1), a hidden semi-Markov model [17], a hidden Markov model without jumps, and a Gated Recurrent Unit [20] neural baseline) on Kolmogorov-Smirnov, Anderson-Darling, Wasserstein-1, and Hellinger pass rates for distributional fidelity, together with the autocorrelation function of absolute growth rates for volatility clustering. Extending the framework to multi-asset workflows required addressing a known limitation of the naive single-index composition: drawing the residual from a per-asset generator inflates the composed marginal variance quadratically in the factor loading. We closed that limitation with a closed-form variance-correction scalar that lets the hybrid HMM marginal enter the factor pipeline while preserving its variance, and we evaluated the full pipeline through a per-ticker one-day Value-at-Risk backtest scored by Kupiec's unconditional-coverage test; an iid-residual diversification-return artifact under daily rebalancing was addressed through a calibrated location-shift bias correction. We found that no single alternative reproduced both quality dimensions simultaneously: GARCH reproduced volatility clustering but failed distributional tests, the standard HMM passed distributional tests but could not generate volatility clustering, and only the hybrid HMM with jumps avoided the failure mode of each alternative.

2 Related Work

Approaches to reproducing the stylized facts of equity markets divide into bottom-up agent-based models, which derive return dynamics from simulated interactions of heterogeneous agents [21–26], and top-down data-driven models, which fit statistical structures directly to observed return series. Agent-based generators produce stylized facts as emergent properties but are demanding to calibrate to specific assets, so the data-driven branch is the focus here. Equity-return generators in that branch have been studied since the geometric Brownian motion baseline of Black-Scholes-Merton [27] failed on the empirical regularities that drive risk and pricing. Mandelbrot [3] showed that price changes carry heavier tails than the Gaussian admits; Fama [28] reviewed the evidence that raw returns have near-zero linear autocorrelation; and Schwert [29] documented the slow decay of absolute-return autocorrelation now known as volatility clustering. The ARCH and GARCH conditional-variance models of Engle [6] and Bollerslev [7] reproduced the clustering but left light-tailed marginals unless fed a fat-tailed innovation, represented neither discrete regimes nor sudden jumps, and controlled persistence through a single decay parameter that cannot separate a calm market that slowly heats up from a sudden crash that lingers [8]. Stochastic-volatility models [30, 9] treat volatility as a latent continuous process at the cost of heavy estimation. Jump-diffusion models [10, 11] inject Poisson events directly into the price path, producing heavy tails by construction and fitting equity markets where announcements and earnings surprises generate large discrete moves [31], but in standard form they offer no mechanism for the volatility persistence that follows a jump.

Hidden Markov models address this gap by embedding regime and jump behavior in a latent discrete chain. Hamilton [32] introduced regime-switching to economics in a two-state form, and later work applied it to volatility forecasting [33] and equity trading [34, 35]. Ryden et al. [16] found that Gaussian-emission HMMs matched heavy tails and the absence of return autocorrelation but failed on volatility clustering; Bulla and Bulla [17] recovered the clustering with hidden semi-Markov models that replaced the geometric dwell-time distribution of every state with a parametric (negative-binomial) alternative. Both rely on the Baum-Welch EM algorithm [18], which is iterative and sensitive to initialization. Deep generative alternatives learn the marginal directly but routinely miss the volatility-clustering signature: Takahashi et al. [12] found that standard GAN architectures produced return sequences that visually resembled financial data yet failed to reproduce the absolute-return autocorrelation function, and Kwon and Lee [14] confirmed the same on TimeGAN [13]. Recent surveys [1, 2, 5] agree that stylized-facts reproduction remains the primary quality criterion, and a pretrained foundation-inference alternative for Markov jump processes [36] has not yet been evaluated on financial data. The hybrid HMM of the present paper picks the latent-state route but follows a different design principle than the HSMM. Rather than replace the dwell-time distribution of every state, it leaves the empirically counted transition matrix in place across the central states and adds an event-driven override that, on a sparse subset of time slices, forces the chain into the tail states for a Poisson-distributed duration. The result is a sparse intervention targeted at tail persistence, with the body of the chain carrying its empirical geometric dwell-time distribution unchanged; combined with a Laplace quantile partition, transitions can be counted directly without EM.

Multi-asset composition layers a cross-sectional structure on top of a univariate generator. The single-index model (SIM) of Sharpe [37] decomposes each asset's growth rate into a market-loaded term plus an idiosyncratic residual, reducing the parameter count relative to a full multivariate fit, and generalizes to multi-factor variants [38, 39] without changing its form. Two residual specifications are common in practice. The full-return specification fits a per-asset generator to the asset's own growth-rate history and plugs the draw in as the residual; it inflates the composed marginal variance quadratically in the factor loading. The residual-fit specification fits the generator to the OLS residual series and composes with no further correction, recovering the factor exactly at the cost of the per-asset generator's distributional structure. Mixture hidden Markov models [40] share latent state across assets but are demanding to estimate in high dimensions; copula approaches [41–43] separate marginals from dependence at the cost of copula-family selection. The variance-corrected composition described in Methods sits between the two residual specifications: a closed-form scalar lets a full-return generator enter the factor pipeline without inflating its marginal variance.

Validation of synthetic financial data combines two families of tests: goodness-of-fit tests for distributional fidelity [44–46] and autocorrelation diagnostics for temporal structure [4]. Recent work has emphasized downstream-task validation as a complement: a synthetic dataset that passes distributional tests can still fail at the task it was generated for. Booth and Fama [47] formalized one such concern: a daily-rebalanced allocator on iid-residual paths extracts a structural *diversification return* that is largely absent in real markets, where residual autocorrelation is small but positive [48]. We adopt the classical distributional metrics together with a downstream Value-at-Risk backtest scored by Kupiec's unconditional-coverage test, and we address the diversification-return artifact through a documented bias correction.

3 Methods

3.1 Hybrid hidden Markov marginal generator

The dataset was 424 United States-listed equities and exchange-traded funds spanning ten years (2014–2024). To validate the hybrid hidden Markov framework on a single representative asset, we focused the headline analysis on the SPDR S&P 500 ETF (SPY), reserving the $2,766$ observations from January 3, 2014 to December 31, 2024 for in-sample fitting and the subsequent 249 trading days of 2025 for out-of-sample testing. The data consisted of daily open, high, low, and close prices and volume-weighted average price values. We computed the excess growth rate $G_{i,j}$ for ticker i between trading days $j-1$ and j from the equity price series $P_{i,j}$ as:

$$G_{i,j} \equiv \left(\frac{1}{\Delta t}\right) \ln\left(\frac{P_{i,j}}{P_{i,j-1}}\right) - r_f, \tag{1}$$

where $\Delta t = 1/252$ is the fraction of a trading year spanned by one trading day, and r_f is a constant continuously compounded risk-free rate derived from STRIPS bond yields. The excess growth rate has units of year^{-1} and is time-additive under continuous compounding.

We defined the hybrid hidden Markov model on this dataset as the tuple $\mathcal{M} = (\mathcal{S}, \mathcal{O}, \mathbf{T}, \mathbf{E}, \bar{\pi})$ [15], where the hidden state $S_t \in \mathcal{S} = \{1, \ldots, N\}$ represented market regimes, the observation G_t at step t was the per-ticker growth rate $G_{i,j}$ of Eq. (1) with the asset index suppressed when only one ticker is in play, the observation space $\mathcal{O} \subset \mathbb{R}$ consisted of continuous excess growth rate values, $\mathbf{T}$ governed regime-to-regime transitions, $\mathbf{E}$ specified the state-conditional emission model, and $\bar{\pi}$ denoted the stationary distribution. States were defined by quantile boundaries of a fitted Laplace distribution: $Q_k = F_L^{-1}(k/N;\, \mu_L, b_L)$, where $F_L^{-1}(\cdot;\, \mu_L, b_L)$ is the inverse cumulative distribution function of the Laplace distribution with location μ_L and scale b_L (both estimated from the in-sample growth-rate history by maximum likelihood), with the endpoints fixed at $Q_0 = -\infty$ and $Q_N = +\infty$. The Laplace was chosen for its sharper peak (better matching the concentration of small price movements [3, 4, 49]) and its closed-form quantile function, which scaled to the 424-ticker pipeline without iterative optimization [19]. An observation was assigned to state k when $Q_{k-1} < G_t \leq Q_k$. The within-state emission followed a location-scale Student-t distribution:

$$G_t \mid S_t = k \;\sim\; \mu_k + \sigma_k \cdot t_\nu, \qquad \nu = 5, \tag{2}$$

where t_ν denotes a standard Student-t random variable with ν degrees of freedom, and μ_k, σ_k are state-conditional location and scale parameters. The value $\nu = 5$ was selected from a sensitivity analysis over $\nu \in \{3, 4, 5, 6, 7, 8, 10, 15, 30, \infty\}$ as the value that minimized the kurtosis gap without degrading distributional fidelity ($\nu = \infty$ recovers the Normal baseline); a head-to-head comparison of Gaussian and Student-t emissions against the HSMM baseline of Bulla and Bulla [17] is given in Table S3 of Sec. E.

With the state space and emission family fixed, we estimated the transition matrix $\mathbf{T}$ through direct frequentist counting: the discrete state assignments allowed straightforward counting of observed transitions between regimes, which avoided the initialization sensitivity of expectation-maximization [19]. A consequence of direct counting was that empirical transition matrices often had low probabilities for exiting central states into tail states, which caused models to lose the effect of volatility clustering too quickly [17]. We corrected this by adding two jump parameters, ϵ representing the probability of a jump event and λ representing the mean jump-event duration, along with a tail-state selection rule. Tail states were defined as $\mathcal{S}_- = \{1, \ldots, N_{\text{tail}}\}$ and $\mathcal{S}_+ = \{N - N_{\text{tail}} + 1, \ldots, N\}$, where N_{tail} was user-configurable. When a jump was triggered, the jump duration K was sampled from a Poisson distribution $K \sim \text{Poisson}(\lambda)$, which is standard for modeling independent events in fixed intervals [50]. During these K steps, the standard Markovian transitions were overridden to target tail states, with a user-configurable bias p_{neg} (default 0.52) that favored negative-tail states to reproduce gain/loss asymmetry (Figure 2). Given the state assignments and transition matrix, the remaining model components were estimated directly from the data: the emission parameters μ_k and σ_k were the sample mean and standard deviation of observations assigned to state k, and the empirical transition matrix exhibited a dominant near-diagonal band reflecting short-range regime persistence, with natural residence times of 1–2 steps in every state including the tails (Fig. S5 in Online Appendix D). The stationary distribution $\bar{\pi}$ satisfied $\bar{\pi} = \bar{\pi}\mathbf{T}$ and was estimated by matrix powering ($\mathbf{T}^{50}$) [51, 52]. An initial state was drawn from $S_0 \sim \bar{\pi}$; at each subsequent time step t the chain either transitioned according to the empirical row $\mathbf{T}_{S_{t-1},\cdot}$ (probability $1 - \epsilon$) or entered a Poisson jump episode (probability ϵ) that forced the state into $\mathcal{S}_-$ or $\mathcal{S}_+$ for K consecutive steps before normal transitions resumed. In both cases an observation was sampled from the emission distribution $G_t \sim \mathbf{E}_{S_t}$. The full simulator is given in Algorithm 1.

With the simulator (Algorithm 1) specified, we calibrated the two jump parameters, the jump probability ϵ and mean jump duration λ [10, 11], via a multi-objective grid search that minimized the discrepancy between historical and simulated market signatures. The objective targeted two stylized facts [4, 16]: volatility clustering, measured as the squared error between the observed and simulated autocorrelation function (ACF) of absolute excess growth rates [29] up to a lag of $L = 25$ trading days (the short-horizon window in which the empirical ACF of $|G_t|$ carries most of its variation; post-fit evaluation in Fig. 3 extends to lag 252 for the visual ACF comparison), and leptokurtosis, captured by a penalty term on the global kurtosis [3]:

$$J(\epsilon, \lambda) = \sum_{\tau=1}^{L} \left(\text{ACF}_{\text{obs}}(\tau) - \overline{\text{ACF}}_{\text{sim}}(\tau)\right)^2 + w_\kappa \left(\kappa_{\text{obs}} - \bar{\kappa}_{\text{sim}}\right)^2, \tag{3}$$

where $\text{ACF}_{\text{obs}}(\tau)$ and κ_{obs} are the empirical absolute-growth autocorrelation at lag τ and the global kurtosis, respectively (kurtosis denoted κ to avoid collision with the jump-duration variable K), and the simulated counterparts $\overline{\text{ACF}}_{\text{sim}}(\tau)$ and $\bar{\kappa}_{\text{sim}}$ were averaged across 200 independent synthetic paths of 2,766 trading days per grid point [50]. We set the kurtosis-penalty weight to $w_\kappa = 0.20$ to balance tail behavior against the temporal ACF term. The grid explored ϵ from 10^{-4} to 2.5×10^{-2} and

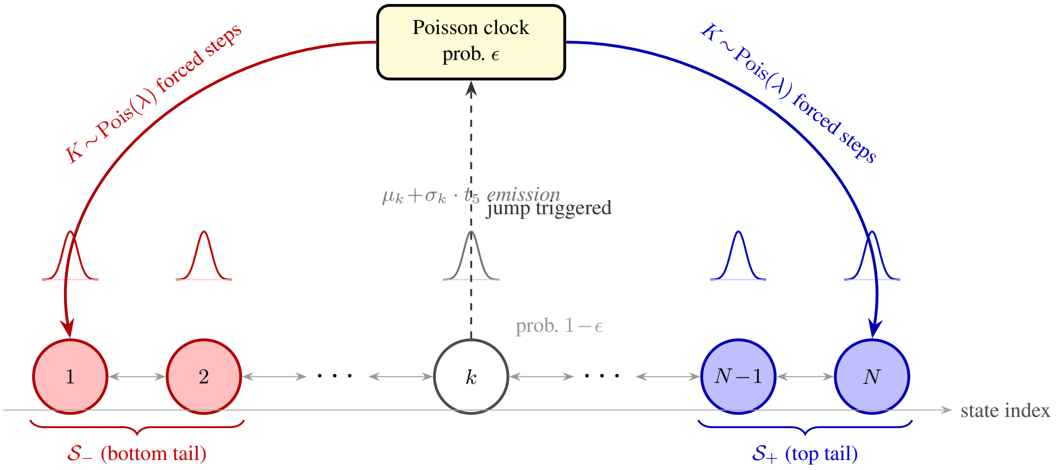

Figure 2: **Architecture of the hybrid hidden Markov model with jump-diffusion (HMM-WJ).** At each step the chain transitions according to the empirical transition matrix $\mathbf{T}$ (probability $1-\epsilon$, thin bidirectional arrows) or enters a Poisson jump episode (probability ϵ, dashed upward arrow to the Poisson clock). During a jump episode the active state is forced to the bottom tail set $\mathcal{S}_-$ (red, lowest-valued states) or the top tail set $\mathcal{S}_+$ (blue, highest-valued states) for $K \sim \text{Poisson}(\lambda)$ consecutive steps before normal Markovian evolution resumes. Small bell curves above each state depict the state-conditional location-scale Student-t ($\nu = 5$) emission distribution. Tail sets are defined as the N_{tail} lowest- and highest-quantile states under the Laplace quantile partition.

Algorithm 1 HMM-WJ simulator.

```
Require: Fitted parameters $(\mathbf{T}, \mathbf{E}, \bar{\pi}, \epsilon, \lambda, p_{\text{neg}}, N_{\text{tail}})$; horizon $T$; tail sets $\mathcal{S}_- = \{1, \ldots, N_{\text{tail}}\}$ and $\mathcal{S}_+ = \{N - N_{\text{tail}} + 1, \ldots, N\}$.
Ensure: State sequence $\{S_t\}_{t=0}^{T}$ and growth-rate path $\{G_t\}_{t=1}^{T}$.
Draw $S_0 \sim \bar{\pi}$;   $k \leftarrow 0$;   $\mathcal{J} \leftarrow \emptyset$        {$k$: forced-step counter}
for $t = 1, \ldots, T$ do
    if $k > 0$ then
        $S_t \sim \text{Uniform}(\mathcal{J})$;   $k \leftarrow k - 1$        {continue forced jump episode}
    else if $\text{Uniform}(0, 1) > \epsilon$ then
        $S_t \sim \mathbf{T}_{S_{t-1},\cdot}$        {standard Markov transition}
    else
        $K \sim \text{Poisson}(\lambda)$;   $k \leftarrow \min(K, T - t)$        {trigger jump of duration $K$}
        With probability $p_{\text{neg}}$ set $\mathcal{J} \leftarrow \mathcal{S}_-$, else $\mathcal{J} \leftarrow \mathcal{S}_+$
        $S_t \sim \text{Uniform}(\mathcal{J})$
    end if
    Draw $G_t \sim \mu_{S_t} + \sigma_{S_t} t_5$        {Eq. (2)}
end for
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λ from 10 to 160 in brute-force evaluation. The HMM-WJ framework produces a per-ticker marginal that reproduces heavy tails, regime persistence, and volatility clustering (validated empirically in Results). Composing these per-ticker marginals into a multi-asset path under a single-index model requires a variance-correction scalar that prevents the naive composition from inflating the per-asset marginal variance.

3.2 Multi-asset extension

The HMM-WJ model produces a per-ticker marginal, but does not capture cross-asset dependence that is required for multi-asset applications. To generalize synthetic path generation beyond a single asset, we composed the per-ticker marginals into a single-index model (SIM) that captured the common market factor and the asset-specific residuals. The Single-Index Model (SIM) [37] decomposes each asset's excess growth rate as:

$$g_i(t) \;=\; \alpha_i \;+\; \beta_i\, g_m(t) \;+\; \varepsilon_i(t), \qquad t = 1, \ldots, T, \tag{4}$$

where $g_m \in \mathbb{R}^T$ is the market growth-rate path with sample variance $\sigma_m^2 = \mathrm{Var}(g_m)$, the pair (α_i, β_i) is the SIM intercept and slope for asset i, and ε_i is the per-asset residual that the construction must specify. Two constraints govern the choice of ε_i: the composed g_i must recover the calibrated (α_i, β_i) under ordinary least squares (OLS), and it must inherit the per-asset marginal the HMM-WJ generator was validated to reproduce (heavy tails, regime structure, jumps).

The naive approach, a *Gaussian-residual composition*, fits a Normal distribution to the OLS residuals of the asset's real history on g_m and samples $\varepsilon_i \sim N(0, \sigma_{\varepsilon,i}^2)$ with $\sigma_{\varepsilon,i}^2 = (1 - R_{i,\mathrm{real}}^2)\,\sigma_i^2$, where $R_{i,\mathrm{real}}^2$ is the OLS coefficient of determination from regressing the asset's history on g_m and σ_i^2 is the empirical variance of the asset's excess growth rates. This approach recovers (α_i, β_i) and pins $\mathrm{Var}(g_i)$ to the empirical σ_i^2, but the composed marginal becomes a mixture of a Gaussian residual and a scaled market path. The asset's heavy tails, regime structure, and jumps, which the per-asset HMM-WJ generator was specifically built to reproduce, are absent from the resulting g_i. Any fixed parametric residual family (Laplace, Student-t) or empirical-bootstrap residual draw inherits the same limitation: pinning the residual to a separate distribution discards the per-asset marginal we already have.

A second option uses the single-asset generator path $\tilde{g}_i$ (with sample variance $\sigma_{\mathrm{gen},i}^2$) directly as the residual:

$$g_i = \alpha_i + \beta_i\, g_m + \tilde{g}_i \tag{5}$$

This approach, which we call *generator-residual composition*, captures the empirical per-asset marginal with variance $\sigma_{\mathrm{gen},i}^2 \approx \sigma_i^2$ calibrated to the asset, so its tails, regimes, and jumps are correct by construction. However, this approach also inflates the marginal variance. To see this, take the variance of both sides of Eq. (5):

$$\begin{aligned}
\mathrm{Var}(g_i) &= \mathrm{Var}(\alpha_i + \beta_i\, g_m + \tilde{g}_i) && \text{(naive composition)} \\
&= \mathrm{Var}(\beta_i\, g_m + \tilde{g}_i) && (\alpha_i \text{ is a constant}) \\
&= \beta_i^2\ \mathrm{Var}(g_m) + 2\beta_i\ \mathrm{Cov}(g_m,\, \tilde{g}_i) + \mathrm{Var}(\tilde{g}_i) && \text{(variance of a sum)} \\
&= \underbrace{\beta_i^2\, \sigma_m^2}_{\text{market}} \;+\; \underbrace{\sigma_{\mathrm{gen},i}^2}_{\text{idiosyncratic}}\ , && (\mathrm{Cov}(\tilde{g}_i, g_m) \approx 0)
\end{aligned} \tag{6}$$

Thus, the variance of g_i exceeds the per-asset (idiosyncratic) target $\mathrm{Var}(g_i) = \sigma_{\mathrm{gen},i}^2$ by exactly $\beta_i^2 \sigma_m^2$, an inflation that grows quadratically in the calibrated beta.

To remove the inflation, we deflate the generator draw before composing: form $\varepsilon_i := s_i\, \tilde{g}_i$ for some scalar $s_i \in (0, 1]$ and require $\mathrm{Var}(g_i) = \sigma_{\mathrm{gen},i}^2$. Under the same independence assumption $\mathrm{Cov}(\tilde{g}_i,\, g_m) \approx 0$, $\mathrm{Var}(\beta_i\, g_m + s_i\, \tilde{g}_i) = \beta_i^2\, \sigma_m^2 + s_i^2\, \sigma_{\mathrm{gen},i}^2$; setting this equal to $\sigma_{\mathrm{gen},i}^2$ and solving yields the variance scale:

$$\boxed{s_i^2 \;=\; 1 \;-\; \frac{\beta_i^2\, \sigma_m^2}{\sigma_{\mathrm{gen},i}^2}} \tag{7}$$

Define the *market loading ratio*:

$$\rho_i \;\equiv\; \frac{\beta_i^2\, \sigma_m^2}{\sigma_{\mathrm{gen},i}^2}, \tag{8}$$

the share of the generator variance consumed by the market component; then $s_i^2 = 1 - \rho_i$, valid whenever $\rho_i < 1$, with the naive composition recovered in the $s_i = 1$ limit. Equation (7) breaks down when $\rho_i \geq 1$: the market loading alone would account for more than the entire generator variance, leaving the idiosyncratic share non-positive, a case that arises for high-β_i assets paired with low-volatility generators or whenever the synthetic market g_m is more volatile than the market on which β_i was calibrated. To handle this case, we imposed a floor $f \in (0, 1)$ on the idiosyncratic share by requiring $s_i^2 \geq f$, and clipped β_i downward whenever the floor was violated. Writing the clip threshold as $\rho_i^{\max} = 1 - f$, the piecewise clip rule is given by:

$$\beta_i^{\mathrm{eff}} \;=\; \begin{cases} \beta_i & \text{if } \rho_i \leq \rho_i^{\max} \\[2ex] \mathrm{sign}(\beta_i)\sqrt{(1-f)\,\dfrac{\sigma_{\mathrm{gen},i}^2}{\sigma_m^2}} & \text{if } \rho_i > \rho_i^{\max} \end{cases} \qquad s_i^2 \;=\; \begin{cases} 1 - \rho_i & \text{if } \rho_i \leq \rho_i^{\max} \\ f & \text{if } \rho_i > \rho_i^{\max} \end{cases} \tag{9}$$

Here $\operatorname{sign}(\beta_i) \in \{-1, +1\}$ is the sign of the calibrated β_i, so the clipped slope inherits the calibrated direction; sign preservation keeps defensive (negative-β) assets defensive even when clipped. The floor f is a free parameter of the construction; in our experiments we use $f = 0.10$, reserving at least 10% of each asset's variance for idiosyncratic noise.

For a typical single stock the heavy tails, regime structure, and jumps live in the marginal that the HMM-WJ generator was built to reproduce. Thus, pinning $\operatorname{Var}(g_i)$ to $\sigma^2_{\text{gen},i}$ keeps the composition aligned with the per-asset calibration. However, index-tracking ETFs reverse the picture: what distinguishes a tracker is its OLS relationship to the market, not its marginal. SPY against itself has $\beta = 1$, $R^2 = 1$, and zero residual variance exactly; QQQ against SPY has $R^2 \approx 0.84$, and SPYG against SPY has $R^2 \approx 0.86$. A variance-preserving composition applied to a tracker supplies a residual whose variance bears no relation to the value $R^2_{i,\text{real}}$ implies, so OLS on the composed path returns $\hat{\beta}$ and $\hat{R}^2$ that drift away from their calibrated values. We therefore branch on the real-data calibration $R^2_{i,\text{real}}$: pick a threshold R^2_{preserve} that separates index-tracking assets from single stocks; in our universe the cumulative distribution sorts itself cleanly, with index ETFs above 0.80 and the most market-like single names below 0.65 (Fig. S3), so $R^2_{\text{preserve}} = 0.80$ captures the trackers without sweeping in single names. For any asset with $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$, inverting the population R^2 identity for Eq. (4) for the residual variance that hits the calibrated R^2 gives (Online Appendix H):

$$\boxed{\sigma^2_{\varepsilon,\text{target}} \;=\; \beta_i^2\, \sigma_m^2 \cdot \frac{1 - R^2_{i,\text{real}}}{R^2_{i,\text{real}}}} \tag{10}$$

The residual that achieves this target variance is the generator path $\tilde{g}_i$ rescaled by the ratio of target to generator standard deviation:

$$\varepsilon_i \;=\; \left(\sqrt{\frac{\sigma^2_{\varepsilon,\text{target}}}{\sigma^2_{\text{gen},i}}}\right) \tilde{g}_i, \tag{11}$$

which can then be copula-reordered and composed via (4). The construction recovers both calibrated quantities by design: $\hat{\beta} \to \beta_i$ in the large-T limit because ε_i is uncorrelated with g_m after the rank reorder, and $R^2 \to R^2_{i,\text{real}}$ by construction. Setting $R^2_{i,\text{real}} = 1$ in Eq. (10) gives $\sigma^2_{\varepsilon,\text{target}} = 0$, so the SPY-against-itself identity $g_i = \alpha_i + \beta_i g_m$ falls out deterministically without special-casing; conversely, when the rescale factor in Eq. (11) exceeds one (which happens when the synthetic market g_m has lower variance than the calibration market) the generator's tails are stretched to fit a target larger than the original draw, and we detect and flag the case so the loss of marginal fidelity is auditable.

The pieces combine into a single procedure (Algorithm 2). For each asset the calibrated $R^2_{i,\text{real}}$ selects a branch: assets with $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$ go through the R^2-preserving construction of Eq. (10), and the rest go through the variance-preserving construction of Eq. (7) with the clip of Eq. (9) engaged when the market loading ratio ρ_i would otherwise exceed $1 - f$. Each asset is labelled with the branch that produced it (R^2-preserving, variance-preserving, or clipped variance-preserving), recorded alongside the output so the path taken per asset can be recovered without external bookkeeping. The optional rank-reorder step in Algorithm 2 accepts any joint dependence structure (empirical copula, t-copula, factor copula); reordering preserves the marginal variance of ε_i, so the variance algebra of Eq. (7) is unaffected and whatever cross-sectional structure is applied to ε_i passes through to g_i unchanged. By construction the composed g_i satisfies three preservation criteria in the large-T limit: *SIM recovery*, where OLS of g_i on g_m returns $\hat{\alpha}_i = \alpha_i$ and $\hat{\beta}_i = \beta_i^{\text{eff}}$ (equal to β_i on the unclipped branch, attenuated by Eq. (9) otherwise); *marginal fidelity*, where $\operatorname{Var}(g_i)$ matches the branch-selected target ($\sigma^2_{\text{gen},i}$ on the variance-preserving branch, or the value implied by the calibrated $R^2_{i,\text{real}}$ on the R^2-preserving branch); and *cross-sectional dependence*, where any joint dependence structure applied to $\{\varepsilon_i\}_{i=1}^N$ passes through unchanged to $\{g_i\}_{i=1}^N$ by the linearity of Eq. (4) in ε_i. The verification algebra, including the tail behavior implied by the floor f, is laid out in Online Appendix H. The construction injects *contemporaneous* market dependence only; lagged dependence (lead-lag effects, market-driven volatility clustering) requires a richer factor structure beyond the scope here.

Algorithm 2 Hybrid SIM composition with variance correction.

```
Require: Market growth-rate path $g_m \in \mathbb{R}^T$ with sample variance $\sigma_m^2$; per-asset generator growth-rate paths $\tilde{g}_i \in \mathbb{R}^T$ for $i = 1, \dots, N$; calibrated SIM parameters $(\alpha_i, \beta_i, R^2_{i,\text{real}})$; floor $f \in (0, 1)$; threshold $R^2_{\text{preserve}} \in (0, 1)$.
Ensure: Composed paths $g_i \in \mathbb{R}^T$ and per-asset construction flags.
for each asset $i = 1, \dots, N$ do
    Compute generator variance $\sigma^2_{\text{gen},i} \leftarrow \text{Var}(\tilde{g}_i)$.
    if $R^2_{i,\text{real}} \geq R^2_{\text{preserve}}$ then
        $\sigma^2_{\varepsilon,\text{target}} \leftarrow \beta_i^2 \sigma_m^2 (1 - R^2_{i,\text{real}}) / R^2_{i,\text{real}}$    {Eq. (10)}
        $\varepsilon_i \leftarrow \left( \sqrt{\sigma^2_{\varepsilon,\text{target}} / \sigma^2_{\text{gen},i}} \right) \tilde{g}_i$
        $\beta_i^{\text{eff}} \leftarrow \beta_i$;   branch $\leftarrow$ $R^2$-preserving
    else
        $\rho_i \leftarrow \beta_i^2 \sigma_m^2 / \sigma^2_{\text{gen},i}$    {Eq. (8)}
        if $\rho_i \leq 1 - f$ then
            $s_i^2 \leftarrow 1 - \rho_i$;   $\beta_i^{\text{eff}} \leftarrow \beta_i$;   branch $\leftarrow$ variance-preserving
        else
            $s_i^2 \leftarrow f$;   $\beta_i^{\text{eff}} \leftarrow \text{sign}(\beta_i) \sqrt{(1 - f)\, \sigma^2_{\text{gen},i} / \sigma_m^2}$;   branch $\leftarrow$ clipped variance-preserving
        end if
        $\varepsilon_i \leftarrow s_i \, \tilde{g}_i$
    end if
    (Optional) apply cross-sectional rank reorder to $\varepsilon_i$.
    $g_i \leftarrow \alpha_i + \beta_i^{\text{eff}} \, g_m + \varepsilon_i$    {Eq. (4)}
end for
```

4 Results

We first measured the empirical SPY data targets that any synthetic generator must reproduce. Descriptive statistics on the real SPY series confirmed all three stylized facts in both the in-sample (2014–2024, $T = 2{,}766$) and out-of-sample (2025, $T = 249$) windows (Table S1): excess kurtosis with Jarque-Bera rejection, persistent volatility clustering, and near-zero linear autocorrelation in raw returns [28].

To meet those empirical targets, we calibrated the jump mechanism of the HMM-WJ simulator (Algorithm 1) via a multi-objective grid search against the observed in-sample SPY series. The grid spanned $\epsilon \in [10^{-4},\, 2.5 \times 10^{-2}]$ and $\lambda \in [10, 160]$ with 200 simulated paths per grid point; we adopted $\epsilon = 10^{-4}$ and $\lambda = 100$ as the working operating point. The composite objective combining ACF and kurtosis targets has multiple comparably-scored local minima at finite Monte-Carlo budgets, so the specific argmin is not the defendable claim; the empirically reproducible claim is that the chosen operating point delivers strong distributional and temporal fidelity on SPY (Table 1). At this operating point, the ACF of $|G_t|$ closely tracked the observed SPY ACF across lags 1–252, indicating that the jump mechanism partially reproduced the empirical ARCH effect. The primary analysis used $N = 100$ states, but a sensitivity study over $N \in \{30, 60, 90, 100, 150, 200\}$ confirmed robustness: KS and AD pass rates remained stable for both HMM-NJ (without jumps) and HMM-WJ (with jumps) across this range, with every state receiving adequate statistical support (Online Appendix F). At $N = 350$, certain states were never visited in the historical record, establishing a practical upper bound on state resolution.

With the model calibrated, we benchmarked HMM-WJ (Algorithm 1) against seven alternative generators on 1,000 synthetic paths of 2,766 trading days each (Table 1, Figure 3). Bootstrap resampling set the non-parametric ceiling (100% KS, 100% coverage); the Gaussian i.i.d. baseline anchored the thin-tailed floor (0% KS). Between these anchors, every parametric and neural baseline excelled on one quality dimension but failed on another. GARCH(1,1) achieved the best temporal fidelity (ACF-MAE 0.031) but a poor distributional fit (5.5% KS). The Gated Recurrent Unit (GRU) [20] closed 83% of the ACF-MAE gap but produced a 0.6% KS pass rate. The hidden semi-Markov model (HSMM) [17] reached 82% KS but its coarse 8-state partition left a 38% kurtosis gap and only

Table 1: **In-sample and out-of-sample model comparison for SPY** ($1{,}000$ simulated paths, significance level $\alpha = 0.05$). Bootstrap: i.i.d. resample of training data. Gaussian/Laplace: i.i.d. draws from MLE-fitted distributions. GARCH: GARCH(1,1) estimated by quasi-maximum likelihood. GRU: 2-layer gated recurrent unit [20] with Gaussian output head. HSMM: hidden semi-Markov model with $K = 8$ Laplace quantile states, negative-binomial dwell times, and Student-t ($\nu = 5$) emissions [17]. HMM-NJ: hidden Markov model ($N = 100$ states) without jumps. HMM-WJ: hybrid model with Poisson jump-duration extension. ACF-MAE: mean absolute error of the ACF of $|G_t|$ over lags 1–252. Wasserstein-1 and Hellinger distance defined in Online Appendix C. Coverage: fraction of 99 empirical quantiles (1st–99th) falling within the [5th, 95th] percentile envelope of the corresponding synthetic quantiles. Standard errors in parentheses: binomial SE for pass rates, $\mathrm{std}/\sqrt{n}$ for kurtosis, Wasserstein-1, and Hellinger; bootstrap SE ($B = 500$) for ACF-MAE and coverage. Best value per row in bold.

Metric	**Bootstrap**	**Gaussian**	**Laplace**	**GARCH(1,1)**	**GRU**	**HSMM**	**HMM-NJ**	**HMM-WJ**
In-sample: 2,766 *trading days* (2014–2024)								
KS pass rate (%) ↑	100.0 (<0.1)	0.0 (<0.1)	44.0 (1.6)	5.5 (0.7)	0.6 (0.2)	82.0 (1.2)	**99.7** (0.2)	97.6 (0.5)
AD pass rate (%) ↑	99.7 (0.2)	0.0 (<0.1)	43.3 (1.6)	1.9 (0.4)	0.2 (0.1)	42.5 (1.6)	**99.1** (0.3)	91.3 (0.9)
Excess kurtosis (observed)	7.715							
Excess kurtosis (simulated) ≈	7.6 (0.08)	−0.0 (<0.01)	3.0 (0.02)	**8.2** (0.37)	5.7 (0.16)	4.8 (0.08)	8.1 (0.14)	7.6 (0.14)
ACF-MAE ↓	0.060 (<0.001)	0.060 (<0.001)	0.060 (<0.001)	**0.031** (0.001)	0.036 (<0.001)	0.059 (<0.001)	0.059 (<0.001)	0.052 (<0.001)
Coverage (%) ↑	100.0 (<0.1)	13.1 (0.1)	37.4 (1.5)	29.3 (1.4)	17.2 (0.5)	68.7 (1.1)	**100.0** (<0.1)	**100.0** (<0.1)
Wasserstein-1 ↓	0.062 (0.001)	0.399 (0.001)	0.138 (0.001)	0.304 (0.006)	0.421 (0.003)	0.176 (0.001)	**0.081** (0.001)	0.101 (0.001)
Hellinger dist ↓	0.042 (<0.001)	0.148 (<0.001)	**0.072** (<0.001)	0.100 (0.001)	0.134 (0.001)	0.113 (<0.001)	0.073 (<0.001)	0.075 (<0.001)
Out-of-sample: 249 *trading days* (2025)								
KS pass rate (%) ↑	97.4 (0.5)	62.2 (1.5)	88.0 (1.0)	80.3 (1.3)	71.8 (1.4)	96.2 (0.6)	**96.7** (0.6)	94.4 (0.7)
AD pass rate (%) ↑	98.5 (0.4)	46.3 (1.6)	92.9 (0.8)	72.0 (1.4)	44.8 (1.6)	**96.7** (0.6)	**96.7** (0.6)	95.1 (0.7)
Excess kurtosis (observed)	6.867							
Excess kurtosis (simulated) ≈	6.0 (0.18)	−0.0 (<0.01)	2.7 (0.05)	1.6 (0.07)	2.9 (0.10)	4.1 (0.13)	**6.4** (0.22)	6.1 (0.13)
ACF-MAE ↓	0.043 (<0.001)	0.043 (<0.001)	0.043 (<0.001)	**0.026** (<0.001)	0.033 (<0.001)	0.042 (<0.001)	0.041 (<0.001)	0.039 (<0.001)
Coverage (%) ↑	100.0 (<0.1)	36.4 (1.9)	74.7 (1.0)	96.0 (1.3)	77.8 (2.9)	**100.0** (0.2)	**100.0** (<0.1)	**100.0** (<0.1)
Wasserstein-1 ↓	0.232 (0.002)	0.452 (0.002)	0.263 (0.002)	0.507 (0.018)	0.488 (0.005)	0.287 (0.002)	**0.258** (0.002)	0.282 (0.006)
Hellinger dist ↓	0.205 (0.001)	0.235 (0.001)	0.211 (0.001)	0.232 (0.001)	0.240 (0.001)	0.239 (0.001)	**0.207** (0.001)	0.210 (0.001)
Parameters estimated	0	2	2	3	37,954	18	2	4

68.7% quantile coverage; a head-to-head isolation of persistence mechanism (HMM-WJ vs HSMM) and emission family (Gaussian vs Student-t) is given in Table S3 (Sec. E). Both HMM variants delivered the highest distributional fidelity among parametric models, with HMM-WJ's pass rates 2–8 percentage points below HMM-NJ's, the cost of jumps occasionally overweighting tail states. The Student-t ($\nu = 5$) emissions closed the kurtosis gap to within 1.3% for HMM-WJ (Normal emissions underestimated by $\sim 29\%$, Table S3). Negative skewness (-0.75 observed, Table S1) was reproduced by both HMM variants, arising from the asymmetric state-occupancy of the Laplace partition rather than any explicit skewness parameter; the symmetric-innovation models (GARCH, Gaussian, Laplace) produced near-zero skewness by construction. On the temporal dimension, HMM-WJ scored 0.052 on ACF-MAE, closing 28% of the gap between the i.i.d. floor (0.060) and the GARCH ceiling (0.031); it was the only non-GARCH non-neural model to reduce ACF-MAE meaningfully below i.i.d., driven by the $\sim 24\%$ of paths that contained at least one Poisson jump and sustained $\mathrm{ACF}(|G_t|)$ decay across all lags (the remaining 76% behaved identically to HMM-NJ, which scored 0.059). Out-of-sample on the held-out 249 days of 2025 the same ranking held (Table 1, Figure S4): HMM-NJ and HMM-WJ achieved 96.7% and 94.4% KS with simulated kurtosis of 6.4 and 6.1 against an observed 6.9, while GARCH's W_1 deteriorated by 67% from in-sample, exposing a tail misspecification that binary pass rates obscured. Cross-asset evaluation on NVDA, JNJ, and JPM produced stable in-sample fits with HMM-WJ KS $\geq 98.9\%$ on all three; the out-of-sample pass rate held at 97.7% for NVDA and attenuated to 74.0 and 79.2% for JNJ and JPM, reflecting larger between-window distributional shifts on the defensive and financial tickers than on either the SPY ETF or the semiconductor name (Table S7, Online Appendix G).

With the per-asset marginals validated on single-asset stylized facts in both windows, we evaluated the multi-asset composition (Algorithm 2) and asked whether those validated marginals composed into a multi-asset construction without breaking either target. We evaluated six composers on a fixed set of per-asset JumpHMM marginals calibrated against the real SPY growth-rate path. This isolated the effect of the composition rule from the effect of market-path randomness: all composers consumed the same real SPY history and the same per-ticker innovation draw, so metric differences across composers reflected the composition rule alone. The ticker universe was a 424-asset US equity/ETF set drawn from Polygon.io daily-close histories, filtered to tickers with a complete in-sample trading history matching the reference length of AAPL ($T = 2{,}767$ daily closes covering January 2014 through December 2024); filtering on the common length preserved 424 tickers, of which the SPY

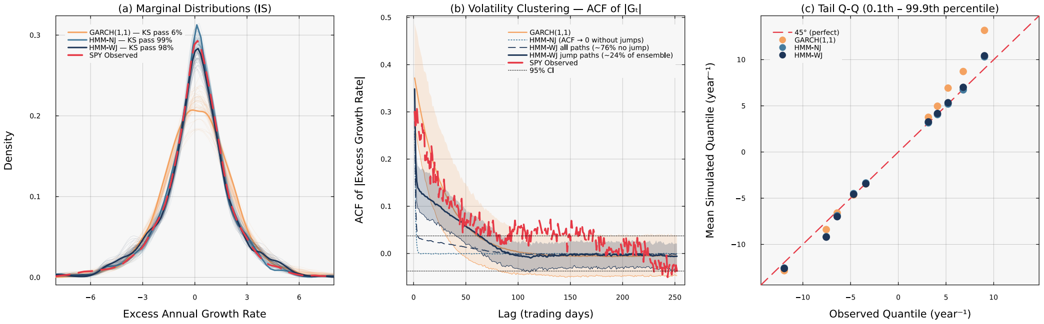

Figure 3: **Head-to-head in-sample model comparison for SPY** ($N = 100$, 1,000 simulated paths). *Panel (a):* marginal density of excess growth rates with in-sample KS pass rates annotated. GARCH(1,1) fails the two-sample KS test on 95% of paths, indicating a systematic shape mismatch with the empirical distribution; both HMM variants pass at rates $\geq 97\%$. *Panel (b):* autocorrelation function of $|G_t|$ at lags 1–252 (shaded bands: 10th–90th percentile across paths). HMM-NJ is structurally incapable of producing persistent volatility clustering; HMM-WJ generates a mixture of two path families, approximately 76% behaving like HMM-NJ and the remaining $\sim 24\%$ containing at least one Poisson jump that sustains $\mathrm{ACF}(|G_t|)$ decay across all lags. *Panel (c):* tail Q-Q plot at the 0.1st–99.9th percentile region; HMM-WJ provides the closest mean quantile match in the extreme tails.

total-return growth-rate series was taken as the market path g_m and the remaining 423 tickers as non-market assets. For each non-market ticker i we fitted a JumpHMM marginal to the asset's own growth-rate history, using the default configuration ($N = 100$ Laplace-partition states, Student-t emissions with $\nu = 5$, annualized excess growth rates with risk-free rate $r_f = 0$ and $\Delta t = 1/252$). The jump mechanism was disabled per ticker by holding the jump probability at $\epsilon = 0$, so the per-ticker generators reduced to the no-jump variant of the framework. The composer test is a marginal-level question (does the composition rule preserve or distort the per-asset marginal under the SIM identity?), and the jump mechanism's effect sits in the temporal-clustering channel rather than the marginal-shape channel; per-ticker tune of (ϵ, λ) would tighten each marginal at the cost of a calibration step per asset, an exchange we made explicit by reporting universe-wide composer results under a shared, jumps-off configuration. For each non-market ticker we then ran OLS against the market path to obtain the calibrated SIM intercept, slope, coefficient of determination, and residual standard deviation. The universe spanned $\beta \in [0.01, 1.79]$ with median $\beta = 1.00$ and quartiles $(0.81, 1.00, 1.18)$, with $R^2_{i,\mathrm{real}}$ ranging from 0.001 to 0.933 (Fig. S3).

A single innovation path drawn from the ticker's JumpHMM marginal was passed unchanged to the *Naive*, *Gaussian SIM*, and *Hybrid* composers (paired innovation protocol), isolating the composition rule from generator stochasticity. Naive plugged the JumpHMM draw into the SIM identity with no correction; Gaussian SIM substituted i.i.d. Gaussian noise scaled to the calibrated residual variance, recovering R^2 exactly but discarding marginal structure; Hybrid applied Algorithm 2 with floor $f = 0.10$ and threshold $R^2_{\mathrm{preserve}} = 0.80$. Three further composers drew innovations from residual generators fit directly to the OLS residual series: *JumpHMM-on-residuals* fitted a fresh JumpHMM marginal via pseudo-price inversion; *Block bootstrap* [53] used Politis–Romano stationary bootstrap replicates with mean block length $\sqrt{T} \approx 50$; *GARCH(1,1)-t* [7] fitted a conditional-variance model per ticker (30 of 423 non-stationary fits excluded from that composer only). The optional cross-sectional rank reorder in Algorithm 2 was skipped so per-ticker metrics reflected the marginal construction alone.

For every composed series we computed three metric families against the ticker's real growth-rate history (metric definitions in Online Appendix C): SIM recovery via OLS on the market path, reporting the absolute deviations $|\hat{\beta} - \beta|$ and $|\hat{R}^2 - R^2_{\mathrm{real}}|$; marginal fidelity via Kolmogorov–Smirnov (KS) and Anderson–Darling (AD) two-sample p-values against the real marginal, plus Wasserstein-1 distance, excess kurtosis, and the Hill upper-tail index on the upper 5% order statistics of $|g|$; and variance preservation via the ratio of composed to generator sample variance. Pass rates were reported at $\alpha = 0.05$, with $R = 100$ replications per ticker yielding aggregates over 42,300 per-ticker observations per composer; the simulation seed was fixed at 1234.

Table 2: **Aggregate scorecard across** 423 **non-market tickers and** 100 **replications per ticker.** Median values per composer. $|\hat{\beta} - \beta|$ and $|\hat{R}^2 - R^2_{\mathrm{real}}|$: absolute deviation between OLS recovery and calibration. KS pass and AD pass: fraction of replications whose two-sample test against the real marginal exceeds $p = 0.05$. W_1: Wasserstein-1 distance to the real marginal. κ: sample excess kurtosis. Hill: tail-index estimate on the upper 5% of $|g|$. All three composers recover β to the same precision; only the hybrid composer preserves the marginal distribution (KS pass 50.4%) while holding the variance close to the generator target.

Composer	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\mathrm{real}}\rvert$	KS pass (%)	AD pass (%)	W_1	κ	Hill
Naive	0.022	0.087	4.5	2.7	0.710	7.767	0.369
Gaussian SIM	0.017	0.008	0.6	0.5	0.682	1.427	0.217
Hybrid	0.018	0.021	50.4	47.6	0.272	7.165	0.365
JumpHMM-on-residuals	0.018	0.015	75.8	70.7	0.232	7.262	0.365
Block bootstrap	0.017	0.020	73.3	66.8	0.232	6.675	0.330
GARCH(1,1)-t	0.017	0.036	67.4	61.4	0.267	6.069	0.317

With the protocol established, we computed per-asset medians of each metric and aggregated across the universe (Table 2). All composers recovered β to within a median absolute error of 0.018 to 0.022, as expected since every composer respects the SIM identity in expectation. The differences between composers appeared everywhere else. The Gaussian SIM baseline recovered R^2 most tightly (median error 0.008) because it targeted R^2 by construction, but it failed marginal fidelity badly: KS pass rate 0.6%, AD pass rate 0.5%, excess kurtosis 1.4 (real data, ~ 13), Hill tail index 0.22 (versus ~ 0.37 on the generator; Fig. S2). The naive composition retained the generator marginal but inflated its variance by $\beta_i^2 \sigma_m^2$, so its KS and AD pass rates sat at 4.5% and 2.7%. The hybrid composer held both properties at once: KS pass rate 50.4%, AD pass rate 47.6%, kurtosis 7.2, Hill 0.36; the R^2 recovery (0.021) came in tighter than the naive baseline but looser than the Gaussian baseline, reflecting that the variance-preserving branch targeted $\sigma^2_{\mathrm{gen},i}$ rather than $R^2_{i,\mathrm{real}}$. The three residual-fit composers sidestepped the variance-inflation problem entirely by fitting a generator directly to the empirical OLS residual series e_i, whose sample variance $\sigma^2_{\varepsilon,\mathrm{real},i}$ excludes the market component by construction; their aggregate KS pass rates were 75.8% (JumpHMM-on-residuals), 73.3% (block bootstrap), and 66.9% (GARCH(1,1)-t), each higher than hybrid's 50.4%. The advantage sat in the body of the distribution rather than the tails: hybrid's median excess kurtosis (7.17) and Hill upper-tail index (0.365) matched JumpHMM-on-residuals (7.26 and 0.365) to within sampling variability, confirming that the variance correction of Eq. (7) preserves the heavy-tailed character of the full-return generator even when the rescaling is applied; block bootstrap and GARCH, by fitting their own residual generators, reproduced the empirical residual series but with slightly lighter tails ($\kappa = 6.68$ and 6.04; Hill 0.330 and 0.318) than either JumpHMM-based recipe.

To test whether the body-fidelity gap propagated into Value-at-Risk accuracy, we ran a per-ticker one-day VaR backtest on the same universe (Table 3). At $\alpha = 0.99$ the nominal exceedance rate is 1%; the naive composer under-breached at 0.67% and the Gaussian baseline over-breached at 1.40%, both failing Kupiec unconditional coverage on the majority of the universe (45 and 47% pass rates), while the hybrid, JumpHMM-on-residuals, block bootstrap, and GARCH(1,1)-t composers all hit coverage within one-tenth of a percentage point of nominal (0.95, 0.99, 1.09, and 1.09% respectively), and their Kupiec pass rates clustered between 72 and 86%. Hybrid and JumpHMM-on-residuals were statistically indistinguishable on this test: the 25 percentage-point body-fidelity gap between them did not translate to a Value-at-Risk accuracy difference. Hybrid's scope was therefore the case where the per-asset generator was specified in full-return units; in that case it delivered tail accuracy on par with a dedicated residual generator without requiring a separate fitting stage.

Having established that body-fidelity did not translate into VaR accuracy, we traced where the KS advantage over the naive baseline came from by plotting per-ticker KS pass rate and per-ticker marginal variance ratio $\mathrm{Var}(g)/\sigma^2_{\mathrm{gen}}$ against the calibrated β (Fig. 4). The mechanism panel made the variance inflation explicit: the naive scatter traced the theoretical curve $1 + \rho_i$, overshooting generator variance by 50–75% at high β, while hybrid sat on the unit line by construction. The KS-pass-rate panel showed the consequence: naive fell from ~ 0.25 at $\beta \approx 0.3$ to below 0.01 past $\beta \approx 0.7$, while hybrid sat between 0.3 and 0.9 across the full range. Applying the branch-selection rule of Algorithm 2 to the 423-ticker universe assigned 421 tickers to the variance-preserving branch, 2 to the

Table 3: **Per-ticker one-day Value-at-Risk backtest across the six composers.** For each (ticker, replication) pair we estimate the VaR threshold at coverage level α from the composed path and count exceedances against the real 2014–2024 growth-rate history. *rate*: mean exceedance rate across tickers (nominal: $1-\alpha$). *SD*: cross-ticker standard deviation of the exceedance rate, in percentage points. *Kupiec pass*: fraction of tickers on which the Kupiec unconditional-coverage test p-value exceeds 0.05. Hybrid and JumpHMM-on-residuals are statistically indistinguishable at both coverage levels; the naive and Gaussian SIM composers fail coverage on a majority of the universe.

	$\alpha = 0.95$			$\alpha = 0.99$		
Composer	rate (%)	SD (pp)	Kupiec pass (%)	rate (%)	SD (pp)	Kupiec pass (%)
Naive	3.55	0.52	10.4	0.67	0.23	45.2
Gaussian SIM	4.06	0.76	42.6	1.40	0.29	47.0
Hybrid	4.75	0.56	80.4	0.95	0.27	79.7
JumpHMM-on-residuals	4.87	0.55	84.0	0.99	0.26	82.9
Block bootstrap	4.78	0.64	76.7	1.09	0.25	86.0
GARCH(1,1)-*t*	4.64	0.93	69.0	1.09	0.35	72.1

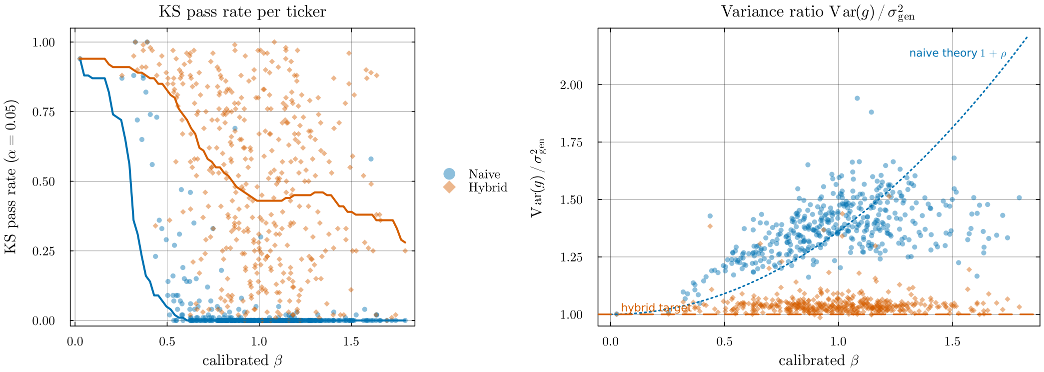

Figure 4: **Hybrid's variance correction and its KS-pass-rate consequence, per ticker vs calibrated β.** Each point is a per-ticker median over 100 replications; thick lines are binned medians. *Left (consequence):* Kolmogorov–Smirnov pass rate against the real marginal at $\alpha = 0.05$. The naive composer (blue) collapses past $\beta \approx 0.6$ as market loading inflates the composed marginal; the hybrid composer (orange) stays between ~ 0.3 and ~ 0.9 across the full β range. *Right (mechanism):* variance ratio $\mathrm{Var}(g)/\sigma^2_{\mathrm{gen}}$. The dotted curve is the theoretical naive inflation $1+\rho$ evaluated at the universe-median σ^2_{gen}; the naive scatter follows it, overshooting generator variance by 50 to 75% at high β. Hybrid sits on the unit line (dashed) by construction.

R^2-preserving branch (QQQ at $R^2 = 0.861$ and SPYG at $R^2 = 0.933$, both passing KS at 99% since the branch recovers both β and R^2 by construction), and none to the clipped variance-preserving branch (Fig. S6, Table S4). The highest-R^2 individual stock (BLK at $R^2 = 0.645$) sat ~ 0.16 below the $R^2_{\mathrm{preserve}} = 0.80$ threshold, so the assignment was empirically tracker-specific on this universe rather than a threshold artifact. Clipping did not trigger anywhere in the universe: $\rho_i = \beta_i^2 \sigma_m^2 / \sigma^2_{\mathrm{gen},i}$ (Eq. (8)) stayed below $1 - f = 0.90$ for every ticker because the JumpHMM marginals carried enough variance to keep the market component below the $f = 0.10$ idiosyncratic floor. Because the R^2-preserving branch had only two empirical assignments, we validated it on a synthetic-tracker grid spanning $(\beta_{\mathrm{true}}, R^2_{\mathrm{true}}) \in \{0.8, 1.0, 1.2\} \times \{0.80, 0.85, 0.90, 0.95, 0.99\}$; the branch selector recovered the target $\hat{\beta}$ and $\hat{R}^2$ within sampling tolerance and passed KS on every cell. Bucketing the universe by β quartile (Table S5), the hybrid KS pass rate declined gently from 67% at low β to 44% at high β, the attenuation noted alongside Algorithm 2: as ρ_i approaches the clipping threshold the market-driven component of g tightens the marginal toward Gaussian along its own axis. Hybrid still beat both baselines by more than an order of magnitude in every bucket. To probe the clipping branch we rescaled the market growth-rate path by $\gamma \in \{1, 2, 3\}$, holding the calibrated marginals fixed (Table S6). At $\gamma = 1$ no ticker crossed the threshold and hybrid KS matched the unstressed value (50.3%); at $\gamma = 2, 3$ the clipping branch activated on 76.4% and 95.2% of tickers and hybrid

KS rose to 71.8% and 77.3%, so clipping is a stability mechanism rather than a degradation under stress.

Finally, we tested the aggregate hybrid KS pass rate of Table 2 against the two hyperparameter choices of Algorithm 2 and against the simulation seed (Fig. S7, Table S8). We re-ran the hybrid composer on the full 423-ticker universe at every point of a 5×5 grid, $R^2_{\text{preserve}} \in \{0.70, 0.75, 0.80, 0.85, 0.90\}$ and $f \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$ (Fig. S7); the hybrid KS pass rate ranged from 50.24% to 50.38% across the 25 cells, a spread of 0.14 percentage points, with the R^2-preserve assignment stable at QQQ and SPYG for every threshold at or below $R^2 = 0.85$ and the floor f not changing any branch assignment in the grid. The aggregate pass rate was therefore invariant to the hyperparameter values within wide tolerances. We then quantified the contribution of generator stochasticity to the aggregate metrics by re-running the full six-composer protocol under 8 random seeds (Table S8). The per-composer KS pass rates held to within fractions of a percentage point: hybrid sat at $50.40 \pm 0.18\%$ across seeds, JumpHMM-on-residuals at $75.61 \pm 0.11\%$, block bootstrap at $73.24 \pm 0.09\%$, GARCH(1,1)-t at $66.96 \pm 0.30\%$, naive at $4.41 \pm 0.07\%$, and Gaussian SIM at $0.60 \pm 0.04\%$. The across-composer differences therefore exceeded the seed-induced spread by roughly two orders of magnitude, and the seed sweep reproduced the relative ordering of the composers without inversions across seeds, confirming that the hybrid-vs-baseline gap and the residual-fit-vs-hybrid body-fidelity gap were not artifacts of the simulation seed.

5 Discussion

The per-asset evaluation positioned HMM-WJ as the only generator in the eight-model comparison to make meaningful progress on distributional and temporal fidelity at once (Table 1; Table S3 for the Gaussian-vs-Student-t emission decomposition). Bootstrap resampling set the non-parametric ceiling on marginal fit but inherited the historical realization wholesale and could not extrapolate beyond it. GARCH(1,1) carried the best temporal score (ACF-MAE 0.031) yet failed the KS test on 95% of in-sample paths, indicating that conditional-variance dynamics on their own did not reproduce the empirical body; the same model's Wasserstein-1 distance to the real marginal deteriorated by 67% from in-sample to out-of-sample, exposing a tail misspecification that binary pass rates had obscured. The Gated Recurrent Unit captured a comparable fraction of the temporal structure with 37,954 estimated parameters but produced a 0.6% KS pass rate. The hidden semi-Markov model passed KS at 82%, but its eight-state quantile partition could not span the full growth-rate range, leaving a 38% kurtosis gap. The jump variant paid a small distributional cost relative to the no-jump variant (2–8 percentage points lower on the KS and AD pass rates in-sample) in exchange for becoming the only non-GARCH non-neural model to reduce ACF-MAE meaningfully below the i.i.d. floor: approximately one path in four contained a Poisson jump that sustained $\mathrm{ACF}(|G_t|)$ decay across all lags, while the remaining 76% behaved like the no-jump variant. Out-of-sample on the held-out 2025 history the same model retained a KS pass rate of 94.4% and excess kurtosis 6.1 against the observed 6.9, with the relative ordering of the eight generators preserved.

Three caveats applied to the per-asset construction. First, the state resolution was sized at $N = 100$ Laplace-partition states by a sensitivity sweep over $\{30, 60, 90, 100, 150, 200\}$, beyond which the count of empirical visits per state thinned, and at $N = 350$ certain states received no support in the historical record; for assets with shorter trading histories or with thinner sampling at the extremes, the practical upper bound on state resolution will be lower, and the partition has to be re-sized rather than carried over directly from the SPY calibration. Second, the jump parameters (ϵ, λ) were tuned once against the in-sample SPY history and applied across the 424-ticker universe; the cross-asset extension in Online Appendix G reports per-ticker tune outputs that produced strong in-sample fits on all three single-stock tickers (KS $\geq 98.9\%$). The composite tune objective has multiple near-equivalent local minima at finite Monte-Carlo budgets, so we do not draw asset-class conclusions from the specific argmin locations; what is reproducible is that the chosen operating point delivers a defensible per-ticker fit, and per-ticker tuning is available as a refinement at the cost of a calibration step per asset. Third, the Student-t emission with $\nu = 5$ closed the kurtosis gap from 29% under Normal emissions to within 1.3% of the observed value, an improvement that came from matching tail mass at the per-state innovation level. The negative skewness reproduced by both HMM variants (-0.75 observed) came from the asymmetric state-occupancy of the Laplace partition rather than from any explicit skewness parameter, so assets whose growth-rate distribution is skewed in a direction the partition cannot induce will not be matched by this construction without a corresponding adjustment to the quantile rule.

We then evaluated the multi-asset extension. Per-asset growth-rate paths were composed under a single-index model (SIM) by drawing each asset's innovation from an independent JumpHMM generator fit to the asset's own full growth-rate history (jumps disabled, $\epsilon = 0$, since the composer test is marginal-level), then rescaling the draw by a closed-form scalar before inserting it into the SIM composition. Six composers were evaluated on a calibrated universe of 424 US equities and exchange-traded funds against the real SPY market path: the naive composition, a Gaussian-residual SIM baseline, the hybrid construction described here, and three residual-fit alternatives (JumpHMM-on-residuals, Politis–Romano block bootstrap [53], and a per-ticker GARCH(1,1)-t fit [7]). The variance correction, together with the sign-preserving clip rule and the R^2-preserving branch, sits between two incomplete baselines: naive composition keeps the generator marginal in the residual draw but distorts it by $\beta^2\sigma_m^2$, while Gaussian-residual SIM recovers β exactly but strips the marginal. The hybrid construction sat between those baselines: it recovered β to the same precision as either, held the generator marginal to its target variance in every ticker, and preserved the generator's tails at a level indistinguishable from the naive composition (Fig. 4, Table 2). The three residual-fit composers produced higher Kolmogorov–Smirnov pass rates than hybrid on marginal body fidelity (67 to 76% versus 50%, Table 2), but matched hybrid on the 99% Value-at-Risk coverage test to within statistical tolerance (Table 3). The two findings together draw the tradeoff: residual-fit composers reach the body of the distribution more tightly by fitting a generator directly to the OLS residual series, at the cost of replacing the full-return JumpHMM marginal with a separately specified residual generator; for tail risk the tradeoff disappears, so the choice between the two approaches is driven by whether the regime-switching and jump structure of the full-return marginal is itself a desired property of the per-asset generator. When it is (for example, when the same generator feeds univariate and multivariate pipelines, or when distributional interpretability of the marginal matters downstream), hybrid delivers 99% VaR accuracy on par with a dedicated residual generator without a separate fitting stage. The variance-inflation gap was large enough to matter in practice: at the high-β end of the universe, naive composition inflated the marginal variance by over 70% relative to the generator target (Fig. 4). Branch selection was driven entirely by the calibrated R^2 and each asset carried a persisted construction flag, so the procedure ran uniformly across the ticker universe without external metadata and left an audit trail.

The evaluation isolated the composition rule by fixing the market path at the real SPY history and using the same per-ticker innovation draw across all three composers. This design yielded a paired comparison of composition rules at the per-ticker level but set aside two questions. The first concerns cross-sectional dependence: we skipped the optional rank-reorder step in Algorithm 2 so that per-ticker metrics reflected the composition rule alone. Rank-reordering preserves the marginal variance of ε_i exactly, so the variance algebra and the per-ticker results here are unaffected by which copula family is layered on top; copula selection is therefore a validation question that is separable from the marginal evaluation reported here, and we defer it to follow-up work. The second concerns market-path uncertainty: the metrics reported here used the real SPY path, and under a simulated market path the composed marginal picks up additional variance from the sampling of $g_m^{(r)}$, which for high-R^2 tickers becomes a noticeable effect. The two deferred questions compose: a reorder-plus-simulated-market pipeline would layer cross-sectional structure on top of per-ticker draws built against sampled market paths, with the variance correction of Eq. (7) still providing the per-asset variance target on each realized $g_m^{(r)}$. A separate scope boundary concerns the downstream use case: the main metrics reported here characterize marginal and factor fidelity of the composition rule on per-ticker targets. To test whether that fidelity translates to a consequential downstream number, we also report a per-ticker one-day 95% and 99% Value-at-Risk backtest of the composed paths against the real 2014–2024 growth-rate histories, with exceedance counts assessed by Kupiec's unconditional coverage test; the per-ticker framing keeps this validation inside the paper's scope, while portfolio-level VaR requires the cross-sectional copula reorder of Algorithm 2 (the optional step we did not run). High-turnover portfolio strategies remain a distinct scope boundary: the temporal structure of the residual draw matters in a way that contemporaneous composition does not capture.

A related boundary on the construction was the clipping rule. Clipping did not trigger anywhere in the US-equity universe used here, because the JumpHMM marginals carried enough variance that $\rho_i = \beta_i^2\sigma_m^2/\sigma_{\text{gen},i}^2$ stayed below 0.90 even for the highest-β ticker. Clipping becomes relevant in two settings we did not evaluate: scenario-driven markets, where the synthetic g_m is constructed to be much more volatile than the calibration market (e.g., stress tests), and universes that mix low-volatility single stocks with extreme betas (e.g., leveraged ETFs). In both settings Algorithm 2 persists the original and effective β_i alongside the output, leaving an audit trail for the attenuation. A related attenuation appeared at the other end of the β spectrum, even when clipping did not trigger: the hybrid KS pass rate fell from 67% at low β to 44% at high β (Table S5), a mild attenuation that followed directly from the composition algebra. As $\beta_i^2\sigma_m^2$ grows toward the idiosyncratic floor $(1-f)\sigma_{\text{gen},i}^2$, the market-driven component carries an increasing share of the marginal, and that component is only as heavy-tailed as g_m. The market's own heavy tails slow the attenuation (SPY growth rates are themselves heavy-tailed after JumpHMM fitting), but the effect is visible. Exact marginal fidelity at high β can be recovered by lowering the floor f (at the cost of less conservative clipping behavior) or by pre-scaling the generator variance so that ρ_i stays small.

The variance correction $s_i^2 = 1 - \rho_i$ relies on $\operatorname{Cov}(\tilde{g}_i, g_m) \approx 0$. The JumpHMM marginals are fit in isolation and sampled independently of the market path, so this holds by construction; if the generator is already correlated with the market (for example, a bootstrap from joint history), the empirical covariance can be subtracted from β_i before applying the correction, or s_i^2 can be estimated directly from the empirical variance of the composed series. The same warning applies to the R^2-preserving branch, whose variance identity assumes ε is uncorrelated with g_m after any reorder. The construction also injects a contemporaneous market factor only: $g_i(t)$ depends on $g_m(t)$ and on $\tilde{g}_i(t)$ but not on $g_m(t-1), g_m(t-2), \ldots$, so lagged market dependence (lead-lag effects, market-driven volatility clustering, asymmetric responses to drawdowns) is not reproduced by this recipe. The natural extension is a lagged-factor SIM, $g_i(t) = \alpha_i + \sum_{k=0}^{K} \beta_{i,k}\, g_m(t-k) + \varepsilon_i(t)$; the variance correction generalizes to that form by replacing $\beta_i^2\sigma_m^2$ with the variance of the full market-factor component, though the empirical value of the extra parameters is beyond the scope here. Lagged idiosyncratic dependence is likewise absent: each generator growth-rate path $\tilde{g}_i$ is sampled independently of the others, so any residual autocorrelation present in the real data is discarded by the composition. This matters most for workflows that evaluate portfolios rebalancing at high frequency. With iid residuals, a rebalancing rule collects the full diversification return of [47], a compounding premium that scales with residual variance and rebalancing frequency and is largely absent in real markets, where daily residual autocorrelation is small but positive [54]. Two generator alternatives in Table 2 preserve temporal structure directly: the block-bootstrap composer resamples blocks of the empirical OLS residual series [53], inheriting whatever autocorrelation is present in the data, and the GARCH(1,1)-t composer fits a conditional-variance model per ticker [7]. Both recover marginal fidelity at the cost of committing the pipeline to a specific residual generator, whereas the variance correction of Eq. (7) is generator-agnostic and accepts a full-return JumpHMM marginal (with its jump and regime structure) as input. When residual autocorrelation is the relevant structure for the downstream task, the bootstrap or GARCH alternatives carry it directly, at the cost of dropping the full-return JumpHMM marginal.

The diversification-return artifact is a property of any SIM construction that draws residuals independently across time: real equity residuals carry a small positive serial autocorrelation [48] that any iid sampler discards, including the JumpHMM marginal used here and the i.i.d. Gaussian and JumpHMM-on-residuals composers of Table 2. The quantitative consequence under daily rebalancing is large enough that absolute-return claims drawn from any of these composers require an explicit correction or an explicit caveat. A 20-asset target-weight allocator deployed against a hybrid-composed 2025 universe produced a median synthetic continuously-compounded growth rate of $\sim 30\%/\text{yr}$, against $11.5\%/\text{yr}$ when the same allocator was deployed against the real 2025 growth-rate history; a static-to-daily-rebalance decomposition on the same paths isolated a $+27.7$ percentage-point jump driven by the rebalance step alone, attributing the ~ 22 pp synthetic-versus-real gap to the diversification return rather than to a calibration error [47, 54, 55] (Online Appendix B). A uniform location-shift correction (Eq. (S14), Online Appendix B) rescales each synthetic wealth path by an exponential drag whose rate is set so the corrected median annualized growth rate matches an external long-run prior, set in our deployment from the prevailing risk-free rate plus an active premium net of frictions [56, 57]. The correction is a per-path monotone transform: it preserves the rank order of paths by terminal wealth, the drawdown shape, and the volatility envelope; only the level shifts. Anchoring to a documented prior rather than to the realized test-year outcome keeps the correction independent of the sample to which the synthetic ensemble is being compared and avoids circular calibration. The realized year then lands inside the corrected ensemble at an honest percentile (in our deployment, real-2025 landed at the 63rd percentile of the corrected ensemble, consistent with 2025 being an above-typical equity year). Online Appendix B details the methodology, contrasts the location-shift correction with a multiplicative haircut alternative that scrambles per-path rank order and distorts the drawdown distribution, and discusses why several intuitive alternatives (parameter drift between calibration and deployment, transaction-cost inflation, AR(1) on residuals without restructuring the copula sampling, residual block-bootstrap) do not address the underlying mechanism within the iid-residual SIM framework. The synthetic data carries the ensemble-shape information the paper validates, percentile structure, drawdown distribution, and relative ranking of strategies on the same paths, both before and after correction; the location-shift correction extends that reliability to absolute Sharpe and CCGR levels under daily rebalancing, at the cost of an external prior.

Four extensions remain open. Per-ticker tuning of the jump parameters (ϵ, λ) on each asset's own history would push the marginal pass rates upward at the cost of a calibration step per asset. Multi-factor and lagged-factor single-index variants replace the single market factor with K factors or with K lags, requiring ρ_i to become the sum of factor contributions and the loading vector to be clipped rather than a scalar, though the branch logic is unchanged; the variance correction generalizes to both forms without changing the algebra, and the empirical evaluation is left to follow-up work. The diversification-return artifact admits a structural fix through autoregressive or block-bootstrap residuals composed with the single-index factor, which requires restructuring the copula sampling step and is the most consequential follow-up for high-turnover downstream workflows. A stress-test harness pairs the clipping rule with a constructed market scenario more volatile than history: enforcing a minimum idiosyncratic share keeps per-asset marginals heavy-tailed in precisely the regime where naive SIM composition breaks down most severely.

6 Conclusion

In this study, we developed a hybrid hidden Markov framework for synthetic equity path generation that discretized continuous excess growth rates into Laplace quantile-defined states and augmented regime switching with a two-parameter Poisson jump-duration mechanism. The Laplace partition lets the transition matrix be estimated by direct frequentist counting, bypassing the Baum-Welch algorithm, and the Poisson override forces the chain to dwell in high-volatility tail states long enough to reproduce empirical clustering. Extending the framework to multi-asset workflows required addressing a known limitation of the naive single-index composition, which inflates the composed marginal variance quadratically in the factor loading; we closed that limitation with a closed-form variance-correction scalar that lets the hybrid HMM marginal enter the factor pipeline while preserving its variance.

On the empirical side, the hybrid HMM marginals passed the Kolmogorov-Smirnov test against the real SPY univariate distribution at 97.6% in-sample and 94.4% out-of-sample, outperforming Bootstrap, Gaussian, Laplace, GARCH, hidden semi-Markov, no-jump HMM, and Gated Recurrent Unit baselines on combined distributional and tail metrics (Table 1). The variance-corrected single-index composer recovered the factor loading to a median absolute error of 0.018 and matched the leading residual-fit alternatives on a per-ticker one-day 99% Value-at-Risk backtest, validating the composed paths against the real 2014–2024 growth-rate histories (Table 2, Table 3). We documented one downstream caveat: an iid-residual diversification-return artifact under daily rebalancing that inflates absolute return levels by roughly 22 percentage points on a deployed 20-asset target-weight allocator (Online Appendix B, Table S2). A uniform location-shift bias correction anchored to a documented long-run prior addresses the artifact as a per-path monotone transform that preserves rank order, drawdown shape, and volatility envelope while shifting only the level.

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Author contributions. A.A. designed the hybrid hidden Markov marginal generator, performed the single-asset experiments and validation framework, and contributed to the manuscript. J.D.V. designed the variance-corrected single-index composition, performed the multi-asset experiments

and downstream Value-at-Risk validation, developed the bias-correction methodology, released the code, and contributed to the manuscript.

Competing interests. The authors declare no competing interests.

Code and data availability. The experiment scripts, evaluation harness (six-composer benchmark, Value-at-Risk backtest, bias-correction utility), and the fitted per-asset marginals are released in the paper repository (`https://github.com/varnerlab/HMM-w-jumps-paper`); the underlying model-fitting library is released separately as `JumpHMM.jl` (`https://github.com/varnerlab/JumpHMM.jl`), pinned as a dependency of the experiment code.

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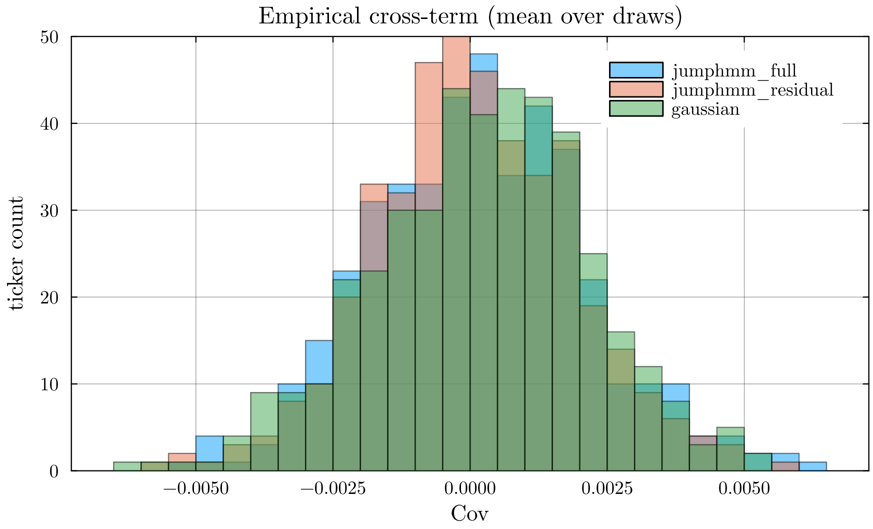

Figure S1: **Empirical normalized cross-covariance** $\mathrm{Cov}(\tilde{g}_i, g_m)/(\sigma_{\mathrm{gen},i}\,\sigma_m)$ **across the** 423**-ticker universe.** Histograms were stacked per generator, with each per-ticker value averaged over $R = 100$ generator growth-rate paths. The full-return JumpHMM, residual-fit JumpHMM, and Gaussian generators all clustered at zero, validating the independence assumption used to derive Eq. (7).

A Empirical cross-term diagnostic

The variance correction of Eq. (7) rests on the assumption $\mathrm{Cov}(\tilde{g}_i, g_m) \approx 0$, which is what allows the derivation to drop the cross-term in the expansion of $\mathrm{Var}(g_i)$. Because the per-asset JumpHMM marginals are fit to each asset's full growth-rate history, including its market-correlated component, the assumption is not automatic: a generator that picked up a residual market loading would invalidate the closed-form scaling. We tested the assumption empirically by sampling $R = 100$ generator growth-rate paths $\tilde{g}_i^{(r)}$ per ticker from each generator and computing the normalized cross-covariance against the real SPY path:

$$\mathrm{cov}_i^{\mathrm{norm}} = \frac{1}{R}\sum_{r=1}^{R} \frac{\mathrm{Cov}(\tilde{g}_i^{(r)},\, g_m)}{\sigma_{\mathrm{gen},i}\,\sigma_m}. \tag{S1}$$

Per-ticker values were plotted as histograms in Fig. S1. Across the 423 tickers, the full-return JumpHMM generator (used by the hybrid and naive composers), the residual-fit JumpHMM generator (used by the JumpHMM-on-residuals composer), and the Gaussian baseline all sat tightly around zero, with no per-generator mean bias outside sampling tolerance (Fig. S1). The independence assumption was therefore satisfied for all three generators, and the derivation of Eq. (7) held in practice.

B Bias correction for daily-rebalanced backtests

This appendix derives the bias correction of Eq. (S14), introduced to close a diversification-return artifact: under daily rebalancing of the SIM-composed synthetic universe, the engine-raw ensemble median sits approximately 22 percentage points above the long-run prior. The variance correction of Eq. (7) is a precondition: it pins the synthetic per-asset second moments to their empirical targets, so the formula value of the diversification return on synthetic paths equals the formula value on real data. The gap is therefore not a structural consequence of the iid-residual draw at the second-order level; it attaches to higher-order Taylor terms, calibration drift, and the documented attenuation of realized diversification returns [54], which the location-shift correction absorbs as a single empirical drag on log-wealth.

The synthetic paths reproduce per-asset marginals and single-factor cross-sectional dependence faithfully but discard temporal structure in the residual draw: at each trading day the per-asset residual $\varepsilon_i(t)$ is sampled independently from the previous day. Real equity residuals carry a small but positive serial autocorrelation [48], typically $\rho \in [0.05, 0.10]$ at the daily horizon, so the iid-residual generator family used here drops a real feature of the data. To make the consequences precise for daily-rebalanced backtests, consider a portfolio of N assets with constant target weights $w_i \geq 0$, $\sum_i w_i = 1$, reset to those weights at the start of each trading day. Throughout this derivation $g_i(t)$ is the annualized log-growth rate of Eq. (1), so the per-period log return on asset i is $g_i(t)\,\Delta t$ with $\Delta t = 1/252$ yr. The wealth process is:

$$W_t \;=\; W_{t-1} \sum_i w_i \exp\bigl(g_i(t)\,\Delta t\bigr), \tag{S2}$$

so the per-step log return on the portfolio is $\log(W_t/W_{t-1}) = \log\sum_i w_i \exp(g_i(t)\,\Delta t)$. We expand this in two stages. A second-order Taylor expansion of the exponential about zero gives $\exp(g_i\,\Delta t) = 1 + g_i\,\Delta t + \frac{1}{2}(g_i\,\Delta t)^2 + O((g\,\Delta t)^3)$. We suppress the time argument inside the derivation, writing g_i for $g_i(t)$. Substituting the Taylor expansion into each summand of $\sum_i w_i \exp(g_i\,\Delta t)$, distributing w_i across the bracketed terms, pulling the i-independent factors Δt and $(\Delta t)^2$ outside the sums, and applying $\sum_i w_i = 1$:

$$\begin{aligned}
\sum_i w_i \exp\bigl(g_i\,\Delta t\bigr) &= \sum_i w_i \bigl[\,1 + g_i\,\Delta t + \tfrac{1}{2}(g_i\,\Delta t)^2 + O\bigl((g\,\Delta t)^3\bigr)\bigr] && \text{(Taylor)}\\
&= \sum_i w_i + \Delta t \sum_i w_i\,g_i + \tfrac{1}{2}(\Delta t)^2 \sum_i w_i\,g_i^2 + O\bigl((g\,\Delta t)^3\bigr) && \text{(distribute)}\\
&= 1 + \Delta t \sum_i w_i\,g_i + \tfrac{1}{2}(\Delta t)^2 \sum_i w_i\,g_i^2 + O\bigl((g\,\Delta t)^3\bigr). && (\textstyle\sum_i w_i = 1)
\end{aligned} \tag{S3}$$

A second-order Taylor expansion of $\log(1+x)$ about $x = 0$ gives $\log(1+x) = x - \frac{1}{2}x^2 + O(x^3)$. Identifying $1 + x$ with the right-hand side of Eq. (S3), the order-Δt correction is $x := \Delta t \sum_i w_i\,g_i + \frac{1}{2}(\Delta t)^2 \sum_i w_i\,g_i^2$, and squaring keeps only the leading $(\Delta t)^2$ term: $x^2 = (\Delta t)^2(\sum_i w_i\,g_i)^2 + O((\Delta t)^3)$, since the cross term carries an extra Δt factor. Substituting:

$$\begin{aligned}
\log\frac{W_t}{W_{t-1}} &= \log\Bigl(1 + x + O\bigl((g\,\Delta t)^3\bigr)\Bigr) && \text{(by Eq. (S3))}\\
&= x - \tfrac{1}{2}\,x^2 + O\bigl(x^3\bigr) && \text{(Taylor)}\\
&= \Delta t \sum_i w_i\,g_i + \tfrac{1}{2}(\Delta t)^2 \sum_i w_i\,g_i^2 - \tfrac{1}{2}(\Delta t)^2\Bigl(\sum_i w_i\,g_i\Bigr)^2 + O\bigl((g\,\Delta t)^3\bigr) && \text{(plug in)}\\
&= \Delta t \sum_i w_i\,g_i + \tfrac{1}{2}(\Delta t)^2 \left(\sum_i w_i\,g_i^2 - \Bigl(\sum_i w_i\,g_i\Bigr)^2\right) + O\bigl((g\,\Delta t)^3\bigr). && \text{(group)}
\end{aligned} \tag{S4}$$

Dividing through by Δt extracts the portfolio's annualized one-step log-growth rate $g_p(t) := \Delta t^{-1}\log(W_t/W_{t-1})$, the portfolio counterpart of the per-asset $g_i(t)$ (still suppressing the time argument):

$$g_p \;=\; \sum_i w_i\,g_i \;+\; \tfrac{1}{2}\,\Delta t \left(\sum_i w_i\,g_i^2 \;-\; \Bigl(\sum_i w_i\,g_i\Bigr)^2\right) \;+\; O\bigl((\Delta t)^2\bigr). \tag{S5}$$

Restoring the time argument, we now take unconditional expectations of Eq. (S5) under stationarity. Define the per-asset annualized mean and variance:

$$\mu_i \;:=\; \mathbb{E}\bigl(g_i(t)\bigr), \qquad \sigma_i^2 \;:=\; \mathrm{Var}\bigl(g_i(t)\bigr), \tag{S6}$$

the per-rate covariance $\Sigma_{ij} := \mathrm{Cov}\bigl(g_i(t), g_j(t)\bigr)$, and the per-rate portfolio variance $\sigma_p^2 := \mathrm{Var}\bigl(\sum_i w_i g_i(t)\bigr) = w^\top \Sigma w$. The $O(\Delta t)$ correction in Eq. (S5) contains two squared quantities, $\sum_i w_i\,g_i(t)^2$ and $(\sum_i w_i\,g_i(t))^2$. Both expand by the second-moment identity, which holds for any random variable X with finite second moment:

$$\mathbb{E}\bigl(X^2\bigr) \;=\; \mathbb{E}\bigl(X\bigr)^2 \;+\; \mathrm{Var}(X). \tag{S7}$$

Applied to $X = g_i(t)$, with $\mathbb{E}(g_i) = \mu_i$ and $\mathrm{Var}(g_i) = \sigma_i^2$, then summed against w_i by linearity of expectation:

$$\begin{aligned}\mathbb{E}\Big(\sum_i w_i\, g_i(t)^2\Big) &= \sum_i w_i\, \mathbb{E}\big(g_i(t)^2\big) && \text{(linearity)} \\ &= \sum_i w_i \big[\, \mu_i^2 + \sigma_i^2 \,\big] && \text{(by Eq. (S7))} \\ &= \sum_i w_i\, \mu_i^2 \;+\; \sum_i w_i\, \sigma_i^2. && \text{(S8)}\end{aligned}$$

Applied to $X = \sum_i w_i g_i(t)$, with $\mathbb{E}(X) = \sum_i w_i\, \mu_i$ by linearity and $\mathrm{Var}(X) = w^\top \Sigma w = \sigma_p^2$ by definition of σ_p^2:

$$\begin{aligned}\mathbb{E}\Big(\big(\textstyle\sum_i w_i g_i(t)\big)^2\Big) &= \mathbb{E}\Big(\textstyle\sum_i w_i g_i(t)\Big)^2 \;+\; \mathrm{Var}\Big(\textstyle\sum_i w_i g_i(t)\Big) && \text{(by Eq. (S7))} \\ &= \Big(\textstyle\sum_i w_i\, \mu_i\Big)^2 \;+\; \sigma_p^2. && \text{(S9)}\end{aligned}$$

Substituting Eqs. (S8)–(S9) into the expectation of Eq. (S5) and grouping the σ_i^2 contributions separately from the μ_i contributions:

$$\mathbb{E}(g_p) \;=\; \sum_i w_i\, \mu_i + \underbrace{\tfrac{1}{2}\,\Delta t\Big(\sum_i w_i\, \sigma_i^2 - \sigma_p^2\Big)}_{D} + \tfrac{1}{2}\,\Delta t\Big(\sum_i w_i\, \mu_i^2 - \Big(\sum_i w_i\, \mu_i\Big)^2\Big) + O\big((\Delta t)^2\big), \tag{S10}$$

the Booth-Fama identity for the daily-rebalanced portfolio in annualized growth-rate units [47, 55]. The middle term, D, is the diversification return: the $O(\Delta t)$ correction equal to half the gap between the weighted-average per-asset variance $\sum_i w_i \sigma_i^2$ and the portfolio variance σ_p^2. The per-rate moments μ_i and σ_i^2 carry units yr^{-1} and yr^{-2} respectively; the products $\Delta t\, \sigma_i^2$ and $\Delta t\, \sigma_p^2$ in Eq. (S10) are the annualized variances (yr^{-1}) that finance practice reports directly. The third term in Eq. (S10) is the cross-sectional dispersion of the per-asset annualized means, of order $\Delta t\, \mu^2 \sim 3 \times 10^{-5}\ \mathrm{yr}^{-1}$ at $\mu \sim 0.08\ \mathrm{yr}^{-1}$ and negligible against the variance term $D \sim \Delta t\, \sigma^2 \sim 4 \times 10^{-2}\ \mathrm{yr}^{-1}$; we drop it. The realized annualized compound growth rate $\bar{g}_p = (T\,\Delta t)^{-1} \log(W_T/W_0) = T^{-1} \sum_t g_p(t)$ converges to $\mathbb{E}(g_p)$ by the ergodic theorem under any stationary residual process, so the long-run limit is fixed by the one-time-slice unconditional joint distribution of $(g_i(t))_i$ and does not, in itself, depend on the serial structure of the residual draw.

To rewrite D in a form that makes its non-negativity manifest, expand $\sigma_p^2 = \sum_i w_i^2 \sigma_i^2 + 2\sum_{i<j} w_i w_j \Sigma_{ij}$ and use the identity $\sum_i w_i(1 - w_i)\sigma_i^2 = \sum_{i<j} w_i w_j (\sigma_i^2 + \sigma_j^2)$ that holds for any weights summing to one:

$$D \;=\; \tfrac{1}{2}\,\Delta t\Big(\sum_i w_i \sigma_i^2 - \sigma_p^2\Big) \;=\; \tfrac{1}{2}\,\Delta t \sum_{i<j} w_i w_j\ \mathrm{Var}(g_i - g_j) \;\geq\; 0, \tag{S11}$$

so D equals one-half the annualized weighted-average pairwise growth-rate-difference variance and is therefore non-negative, with equality only when every pair (i, j) comoves perfectly ($\mathrm{Var}(g_i - g_j) = 0$) or all weight concentrates on a single asset. The mechanism is the rebalancing rule itself: weights drift between rebalances, the rebalancer sells whatever has outperformed (its weight rose above target) and buys whatever has underperformed (its weight fell below target), and the trade systematically converts period-by-period cross-sectional dispersion into compounded growth. Markowitz [55] described this as the variance bonus of the long-run mean-variance allocation; Booth and Fama [47] called it the diversification return.

Substituting the single-index decomposition $g_i(t) = \alpha_i + \beta_i\, g_m(t) + \varepsilon_i(t)$, with market variance $\sigma_m^2 := \mathrm{Var}(g_m(t))$, per-asset residual variance $\sigma_{\varepsilon,i}^2 := \mathrm{Var}(\varepsilon_i(t))$, and $\mathrm{Cov}(g_m, \varepsilon_i) = 0$ (verified in Sec. A), into Eq. (S11) gives:

$$\mathrm{Var}(g_i - g_j) \;=\; (\beta_i - \beta_j)^2 \sigma_m^2 \;+\; \mathrm{Var}(\varepsilon_i - \varepsilon_j), \tag{S12}$$

Table S1: **Descriptive statistics and stylized-facts tests for SPY daily excess growth rates.** In-sample: 2014–2024 ($T = 2{,}766$); out-of-sample: 2025 ($T = 249$). Gaussian and Laplace columns show maximum likelihood estimation (MLE) fits to the in-sample data. JB: Jarque-Bera normality test. LB: Ljung-Box autocorrelation test at lag 20. * denotes rejection at $\alpha = 0.05$.

Statistic	**IS Observed**	**OoS Observed**	**Gaussian**	**Laplace**
Mean (annualized, %)	6.31	11.60	6.31	14.19
Std Dev (annualized, %)	214.50	220.62	214.46	205.93
Skewness	-0.753	-1.093	0	0
Excess Kurtosis	7.715	6.867	0	3
JB normality test	reject* ($p < 0.001$)	reject* ($p < 0.001$)	n/a	n/a
LB test on G_t (lag 20)	reject* ($p < 0.001$)	reject* ($p = 0.019$)	n/a	n/a
LB test on $\lvert G_t\rvert$ (lag 20)	reject* ($p < 0.001$)	reject* ($p < 0.001$)	n/a	n/a

since g_m and the residuals are uncorrelated, so the diversification return splits cleanly into a market-beta-dispersion component and an idiosyncratic component:

$$D \;=\; \underbrace{\tfrac{1}{2}\,\Delta t\,\sigma_m^2 \sum_{i<j} w_i w_j (\beta_i - \beta_j)^2}_{D_{\mathrm{mkt}}} \;+\; \underbrace{\tfrac{1}{2}\,\Delta t \sum_{i<j} w_i w_j \;\mathrm{Var}(\varepsilon_i - \varepsilon_j)}_{D_{\varepsilon}}, \tag{S13}$$

where the market-beta-dispersion component D_{mkt} equals $\frac{1}{2}\,\Delta t\,\sigma_m^2[\sum_i w_i\beta_i^2 - (\sum_i w_i\beta_i)^2]$, the weighted cross-sectional β dispersion scaled by the annualized market variance $\Delta t\,\sigma_m^2$. Across the 423-ticker universe the empirical β distribution spans $[0.01, 1.79]$ with interquartile range $[0.81, 1.18]$ and the SPY annualized variance $\Delta t\,\sigma_m^2 \approx 0.04\ \mathrm{yr}^{-1}$, so D_{mkt} contributes at most a few percentage points per year for any diversified equity portfolio. The idiosyncratic component D_ε is the dominant term: with residuals approximately orthogonal across assets at the same time slice, $\mathrm{Var}(\varepsilon_i - \varepsilon_j) \approx \sigma_{\varepsilon,i}^2 + \sigma_{\varepsilon,j}^2$, and D_ε scales with the weighted-average annualized idiosyncratic variance $\Delta t\,\sigma_{\varepsilon,i}^2$ per name. Concentrated allocators on high-idiosyncratic-volatility names therefore inherit large formula values of D_ε structurally.

The iid construction of the residual draw fixes both terms of Eq. (S13). The variance correction of Eq. (7) pins the synthetic per-asset variance $\sigma_{\mathrm{gen},i}^2 = \beta_i^2\sigma_m^2 + \sigma_{\varepsilon,i}^2$ to its empirical target, so the formula value $D = D_{\mathrm{mkt}} + D_\varepsilon$ on synthetic paths equals the formula value under any alternative residual generator matched on the same unconditional second moments, including a stationary AR(1) residual generator $\varepsilon_i(t) = \phi\,\varepsilon_i(t-1) + \eta_i(t)$ with iid innovations η_i tuned so that $\mathrm{Var}(\varepsilon_i) = \sigma_{\eta,i}^2/(1-\phi^2) = \sigma_{\varepsilon,i}^2$. The AR(1) parameter ϕ leaves the unconditional joint distribution of $(g_i(t))_i$ at a single time slice unchanged, which by Eqs. (S10), (S13) means ϕ enters at zeroth order in $\mathbb{E}(g_p)$ and at first order only in the long-run variance of the sample mean ($\mathrm{Var}(\bar{g}_p) \to T^{-1}\,\sigma_p^2\,(1+\phi)/(1-\phi)$). The synthetic-ensemble median across 1,000 paths therefore converges to the same formula value whether the residuals are drawn iid or with positive serial autocorrelation: the iid versus AR(1) distinction shows up in per-path sampling noise, not in the ensemble median itself. The empirical excess of the engine-raw median over an external long-run growth prior, ~ 22 percentage points, is therefore not a structural artifact of the iid sampling at the level of the second-order formula. It is the combined empirical magnitude of the higher-moment corrections to the Taylor expansion (heavy-tailed JumpHMM marginals carry non-trivial third and fourth cumulants), the time-varying volatility and correlation across the 2014–2024 calibration window relative to a long-run prior, and the documented attenuation of realized diversification returns on real equity portfolios at all rebalancing horizons [54]. The location-shift correction of Eq. (S14) treats this composite shortfall as a single empirical drag b to be absorbed and is agnostic to its decomposition.

To anchor the scale of the artifact, we deployed a 20-asset long-only target-weight allocator (calibration window 2014–2024, deployment year 2025, names with calibrated (α, β) and $\max\beta = 1.21$, 5 basis-point round-trip cost) against the hybrid-composed universe of the main text and decomposed the realized path-mean annualized growth rate $\bar{g}_p \equiv (T\,\Delta t)^{-1}\log(W_{p,T}/W_{p,0})$ into static and rebalancing-driven components (Table S2). The static-to-daily-rebalance switch on the same target weights produced the $+27.73$ percentage-point jump that isolates the diversification-return term of Eq. (S10); adding a signal on top of the daily rebalance moved the median by a further $+0.13$ pp, immaterial compared with the rebalancing-driven bonus. The same allocator on the real 2025 growth-

Table S2: **Diversification-return decomposition on a 20-asset target-weight allocator deployed against the hybrid-composed 2025 universe.** Median path-mean annualized growth rate $\bar{g}_p$ across the synthetic ensemble. Equal-weight buy-and-hold serves as a passive baseline; the target-weight allocation is reported under static (no intra-year rebalance) and daily-rebalanced regimes, with and without a signal on top of the target weights. The static-to-daily jump of $+27.73$ pp on the same target weights isolates the diversification return of Eq. (S13). The same allocator deployed against the real 2025 growth-rate history is shown for comparison.

Allocation rule (synthetic 2025 universe)	$\bar{g}_p$ **(%/yr)**
Equal-weight buy-and-hold	7.68
Target-weight, static (no rebalance)	1.97
Target-weight, daily rebalanced	29.69
Target-weight, daily rebalanced + signal	29.83
Real 2025 history, same allocator	11.50

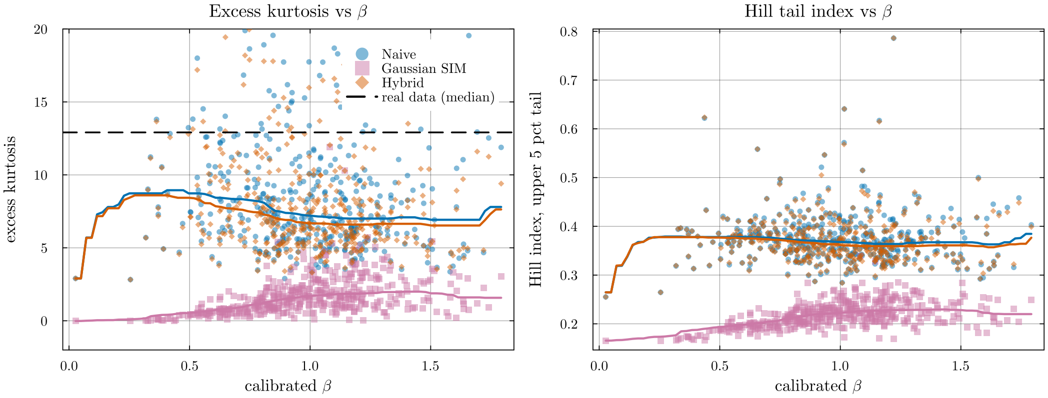

Figure S2: **Heavy-tail preservation: excess kurtosis and Hill tail index vs calibrated β.** Each point is a per-ticker median over 100 replications. *Left:* excess kurtosis, y-axis clipped to $[-2, 20]$ for readability. The dashed line marks the median real-data excess kurtosis across the universe ($\kappa_{\text{real}} \approx 13$); the hybrid and naive composers (orange and blue) track the generator's kurtosis, while Gaussian SIM collapses to $\kappa \approx 1$. *Right:* Hill tail-index estimator on the upper 5% of $|g|$. The hybrid and naive medians sit near 0.37 throughout the β range; Gaussian SIM drops to ~ 0.22, indicating the shift toward a thinner-tailed distribution. The hybrid's median sitting below κ_{real} reflects the JumpHMM marginal's own kurtosis rather than a composition-rule distortion: hybrid tracks naive (inheriting generator tails) while Gaussian SIM does not.

rate history produced $11.5\%/\text{yr}$, leaving the synthetic median running ~ 22 percentage points above reality. The decomposition makes clear that the gap is a structural feature of the iid-residual generator family rather than a calibration error: the allocator choice, the cost model, the marginal calibration, and the cross-sectional structure all behave on synthetic paths exactly as they behave on real data; only the temporal structure of the residual draw differs.

Because the artifact acts as a uniform multiplicative drag on the path in log-wealth space rather than a distortion of the path's shape, a uniform location-shift correction in the same space restores the median absolute level without disturbing the rank order, the drawdown geometry, or the volatility envelope of the synthetic ensemble. Multiplying each path-p wealth trajectory $W_{p,t}$ at integer step t by a uniform exponential drag at rate b that closes the gap between the engine's median annualized growth rate and an external long-run prior $\bar{g}^{\text{prior}}$:

$$W^{\text{corr}}_{p,t} \;=\; W_{p,t} \cdot \exp\!\big(-\, b \cdot t\,\Delta t\big), \qquad b \;=\; \operatorname*{median}_{p} \bar{g}_p \;-\; \bar{g}^{\text{prior}}, \tag{S14}$$

with Δt the integration step in years and b, $\bar{g}_p$, $\bar{g}^{\text{prior}}$ all in the same per-year growth-rate units, accomplishes exactly this. Working in log-space rather than additively keeps the correction multiplicative on wealth, which is the natural scale for compounding returns and preserves drawdown geometry period-by-period; the transform is monotone in $W_{p,t}$ for every path and every step, so per-path rank

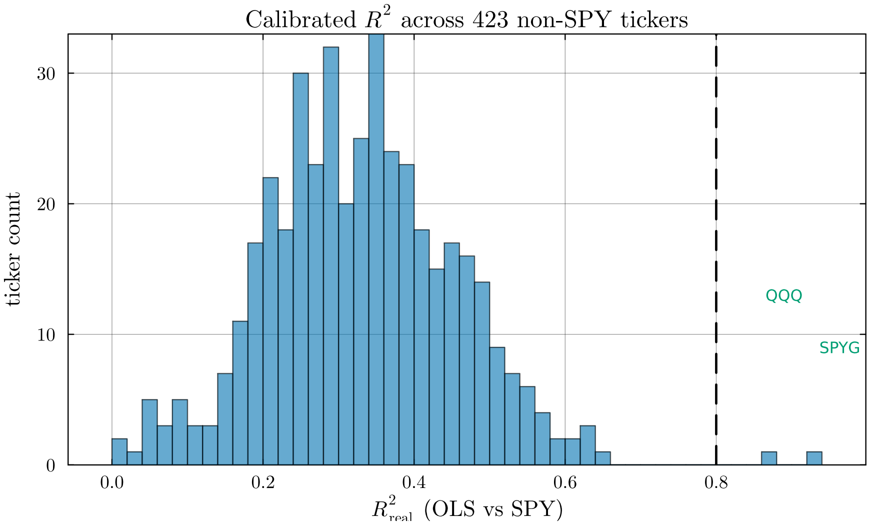

Figure S3: **Calibrated R^2 distribution across the 423 non-SPY tickers.** Histogram of OLS R^2 against SPY over the 2014–2024 window. The two tickers crossing the $R^2_{\text{preserve}} = 0.80$ threshold (dashed line) are index ETFs (QQQ, SPYG); the highest-R^2 individual stock (BLK, 0.645) sits ~ 0.16 below the threshold. The R^2-preserving branch's $n = 2$ occupancy is therefore a property of the single-factor SIM on S&P 500 names against SPY, not a sample-size artifact or a data-peeked threshold.

order on terminal wealth is preserved exactly. The prior $\bar{g}^{\text{prior}}$ is set from external long-run data and is typically near $8\%/\text{yr}$ at the time of writing, decomposable into the prevailing risk-free rate (roughly $4.5\%/\text{yr}$ on a five-year Treasury basis), the historical long-run equity premium of 4–6 pp [56, 57], less 1–2 pp for fees, turnover, and market-impact frictions, leaving ~ 3.5 pp of active premium net of costs. Anchoring to a documented prior rather than to the realized test-year outcome avoids circular calibration: anchoring to the realized year would force the synthetic median to collapse onto the test sample by construction and would misrepresent typical years (in our deployment, the realized 2025 was an above-typical equity year, so anchoring to its $11.5\%/\text{yr}$ outcome would have suppressed the corrected ensemble below its appropriate long-run level). We verified the anchor on a smoke test: the corrected ensemble matched the prior at its median by construction, and the real-2025 outcome landed at the 63rd percentile of the corrected ensemble, consistent with 2025 being an above-typical equity year.

Several alternative fixes were considered and rejected. A path-filtering rule that drops synthetic paths above a $\bar{g}_p$ threshold turns the threshold itself into a tuning target and removes only a subset of the artifact's consequences while leaving the structural artifact intact. Inflating per-trade transaction costs to absorb the bonus would require round-trip costs of ~ 70 basis points versus the realistic 1–10 basis points seen on liquid US equity, shifting the misrepresentation from the synthetic generator to its cost model. Drifting the calibrated parameters between calibration and deployment to widen the predictive envelope, with drift scale varied across $\{0, 1, 2, 3, 5\}$ multiples of the OLS standard error, moved the median Sharpe ratio by less than 0.01 across the sweep, confirming that β is estimated too tightly for SE-scale drift to absorb the artifact. Imposing an AR(1) on the residual draw inside the existing pipeline would conflict with the copula reorder step, which destroys temporal structure by construction; recovering the AR(1) requires a block-ordered copula sampler, which is a structural redesign rather than a downstream correction. The block bootstrap of empirical residuals [53] preserves both temporal and cross-sectional structure of the residual series by construction and is a valid alternative generator pipeline, listed as a future-work item that replaces the JumpHMM marginal rather than correcting it. A multiplicative haircut on the engine-versus-static deviation, $W^{\text{corr}} = W_{\text{static}} \cdot (W_{\text{engine}}/W_{\text{static}})^k$, hits the same anchor at k chosen to match the prior, but its per-path rank correlation with the raw engine drops well below one and the corrected drawdown distribution inherits static-allocation geometry (drawdowns widen materially relative to the raw engine).

The location-shift correction of Eq. (S14), by contrast, preserves these properties: the corrected ensemble carries the ensemble-shape information the paper validates, percentile diagnostics overlaying real outcomes against the synthetic distribution, drawdown distribution and tail-shape analysis, and relative ranking of strategies evaluated on the same corrected paths. It does not remove the underlying iid-residual limitation, so the residual autocorrelation structure itself, absolute Sharpe under daily rebalancing, and the choice of rebalancing frequency remain outside the scope of the correction; the structural fix for those targets is the AR(1) or GARCH residual structure restructured around the copula reorder, listed as future work.

C Validation metric definitions

The distributional and tail metrics reported in the main results were defined as follows. The two-sample Kolmogorov-Smirnov statistic between the empirical CDFs of the observed and simulated growth-rate samples (G_{obs} of length n and G_{sim} of length m) is $D = \sup_x |F_n(x) - F_m(x)|$ [44, 45]; the test rejects when D exceeds the critical value at level $\alpha = 0.05$. The Anderson-Darling statistic places greater weight on the tails via the integral $A^2 = nm/(n+m) \int [F_n(x) - F_m(x)]^2 / [F(x)(1 - F(x))] \, dF(x)$, where F is the pooled empirical CDF [46]. The Wasserstein-1 distance between two empirical distributions of equal length is the average of sorted-quantile differences:

$$W_1 = \frac{1}{n} \sum_{i=1}^{n} \left| G_{\mathrm{obs}}^{(i)} - G_{\mathrm{sim}}^{(i)} \right|, \tag{S15}$$

with $G^{(i)}$ the ith order statistic. The Hill upper-tail index is the classical order-statistic estimator:

$$\hat{\xi} = \frac{1}{k-1} \sum_{i=1}^{k-1} \log\left(\frac{X_{(i)}}{X_{(k)}} \right), \tag{S16}$$

with $X_{(1)} \geq X_{(2)} \geq \ldots$ the ordered upper-tail absolute returns; k is set to the upper 5% of the sample. For a Pareto-tailed distribution with shape α, $\hat{\xi} \approx 1/\alpha$. The Hellinger distance between two density estimates p and q on a common bin grid is:

$$H(p, q) = \frac{1}{\sqrt{2}} \left\| \sqrt{p} - \sqrt{q} \right\|_2 \; \in \; [0, 1], \tag{S17}$$

with $H = 0$ for identical densities and $H = 1$ for disjoint supports.

D Transition matrix internals and partition diagnostics

We inspected the three internal components of the fitted HMM-WJ on SPY with $N = 100$ states (Fig. S5). The fitted Laplace cumulative distribution function overlays the empirical CDF closely, with the 99 vertical lines marking the equal-probability quantile boundaries Q_k that define the hidden states, confirming the Laplace fit as an accurate marginal model for the SPY excess growth-rate series and validating the partition rule of Eq. (1). Plotting the estimated transition matrix $\mathbf{T}$ in $\log_{10}$ color scale, a dominant near-diagonal band reflects short-range regime persistence, with most off-diagonal mass concentrated within ± 5 states of the diagonal. The expected natural residence time $1/(1 - T_{kk})$ for each state on a $\log_{10}$ scale, with the bottom tail set $\mathcal{S}_-$ (states 1–5, red) and the top tail set $\mathcal{S}_+$ (states 96–100, teal) highlighted, makes the gap to the empirical clustering scale explicit. Under pure Markovian dynamics every state, including the tail states, exits within 1–2 steps on average, far too quickly to reproduce empirical volatility clustering; the dashed line marks the operating-point mean jump duration $\lambda = 100$ steps and quantifies the two-order-of-magnitude gap between natural residence times and the Poisson jump override. Quantile-bin occupancy was approximately uniform over the in-sample window, consistent with the Laplace fit's calibration to the empirical CDF; the residual sample-size variation across bins was smaller than the within-bin emission variance, so all 100 states received adequate statistical support.

E Emission distribution and semi-Markov comparison

To isolate the contributions of the emission family and the persistence mechanism, we compared three variants on SPY (Table S3): the HSMM baseline of Bulla and Bulla [17] ($K = 8$ states,

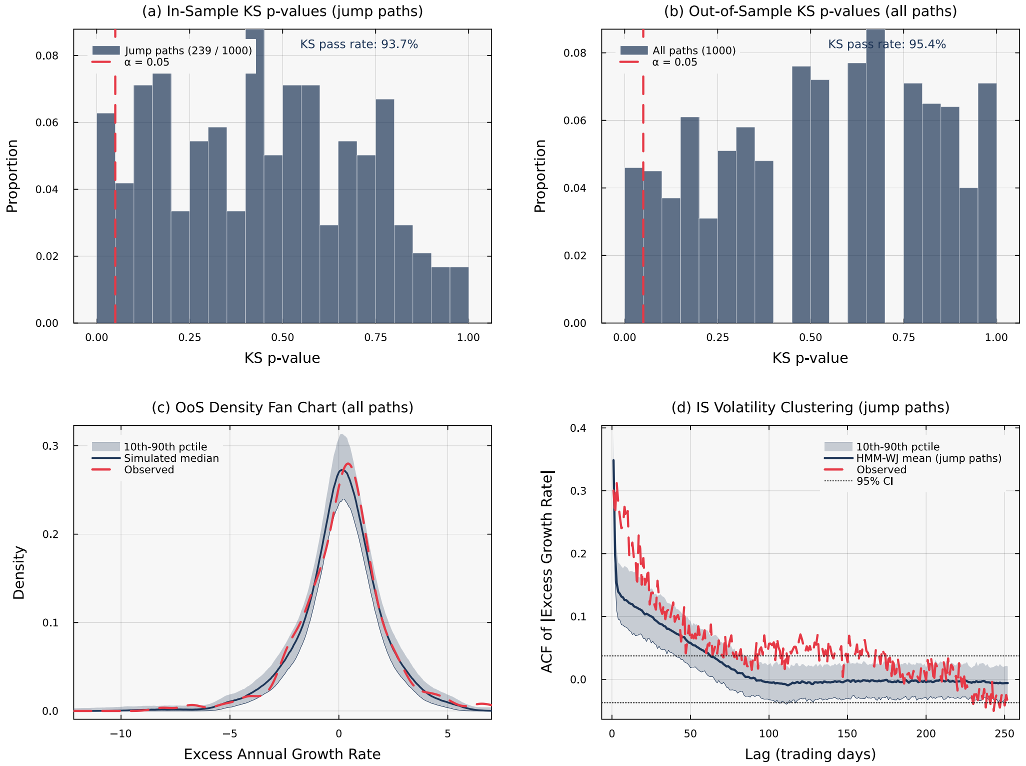

Figure S4: **Statistical validation of HMM-WJ** ($N = 100$, $\alpha = 0.05$). Panels (a) and (d) condition on the $\sim 24\%$ of in-sample paths containing jump events. (a) In-sample KS p-values for 239 jump paths (pass rate 93.7%). (b) Out-of-sample KS p-values for all $1{,}000$ paths (pass rate 95.4%). (c) Out-of-sample density fan chart; the observed density falls within the simulation envelope. (d) In-sample ACF of $|G_t|$ for jump paths; the mean closely tracks the observed ACF.

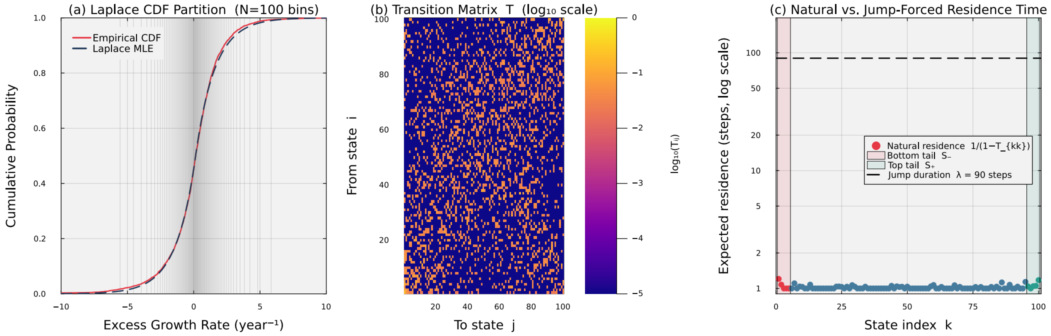

Figure S5: **Fitted HMM-WJ internals for SPY with $N = 100$ states.** Panel (a) overlays the fitted Laplace CDF on the empirical CDF; the 99 vertical lines mark the equal-probability quantile boundaries Q_k that define the hidden states. Panel (b) displays the estimated transition matrix $\mathbf{T}$ in $\log_{10}$ color scale; the dominant near-diagonal band reflects short-range regime persistence. Panel (c) plots the expected natural residence time $1/(1 - T_{kk})$ for each state on a $\log_{10}$ scale, with the bottom tail set $\mathcal{S}_-$ (states 1–5, red) and the top tail set $\mathcal{S}_+$ (states 96–100, teal) highlighted. The dashed horizontal line marks the operating-point mean jump duration $\lambda = 100$ steps, quantifying the two-order-of-magnitude gap between natural residence times and the jump-duration override.

Table S3: **Three-way comparison isolating the emission distribution and the persistence mechanism for SPY.** Standard errors in parentheses; bold marks the best value per row. HSMM: hidden semi-Markov model ($K = 8$ states, negative-binomial dwell times, Student-t emissions) [17]; HMM-WJ: hybrid hidden Markov model with jump-duration override ($N = 100$). All entries are computed from 1,000 simulated paths at significance level $\alpha = 0.05$.

Metric	**HSMM (Student-t)**	**HMM-WJ (Gaussian)**	**HMM-WJ (Student-t)**
States	$K = 8$	$N = 100$	$N = 100$
Parameters	18	4	4
In-sample (2,766 trading days, 2014–2024)			
KS pass rate (%)	82.0 (1.2)	97.0 (0.5)	**97.6** (0.5)
AD pass rate (%)	42.5 (1.6)	90.5 (0.9)	**91.3** (0.9)
Excess kurtosis	4.8 (0.08)	5.5 (0.03)	**7.6** (0.14)
ACF-MAE	0.059 (<0.001)	**0.052** (<0.001)	**0.052** (<0.001)
Wasserstein-1	0.176 (0.001)	**0.101** (0.002)	**0.101** (0.001)
Hellinger	0.113 (<0.001)	0.076 (<0.001)	**0.075** (<0.001)
Out-of-sample (249 trading days, 2025)			
KS pass rate (%)	**96.2** (0.6)	95.0 (0.7)	94.4 (0.7)
AD pass rate (%)	**96.7** (0.6)	95.9 (0.6)	95.1 (0.7)
Excess kurtosis	4.1 (0.13)	4.9 (0.08)	**6.1** (0.13)
ACF-MAE	0.042 (<0.001)	0.040 (<0.001)	**0.039** (<0.001)
Wasserstein-1	0.287 (0.002)	**0.275** (0.005)	0.282 (0.006)
Hellinger	0.239 (0.001)	0.212 (0.001)	**0.210** (0.001)

Table S4: **Hybrid composer broken out by branch.** N_{tickers}: number of tickers assigned to each branch by the $R^2_{\text{preserve}} = 0.80$ threshold. The clipped variance-preserving branch does not trigger anywhere in the universe ($\rho_i < 0.90$ for all tickers) and is omitted. The two trackers in the R^2-preserving branch (QQQ and SPYG) attain a KS pass rate of 99%, consistent with the branch recovering both β and R^2 by construction.

Branch	N_{tickers}	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	W_1	κ
`hybrid`	421	0.018	0.021	50.1	0.273	7.142
`r2-preserve`	2	0.005	0.001	99.0	0.109	11.525

negative-binomial dwell times, Student-t emissions), HMM-WJ with the original Gaussian emissions ($N = 100$), and HMM-WJ with Student-t emissions ($N = 100$). The HMM-WJ variants exceeded the HSMM on in-sample distributional fidelity by 15 percentage points on KS and 49 percentage points on AD regardless of emission choice, attributing that gap to the fine-grained quantile partition over the coarse $K = 8$ HSMM partition. Holding the HMM-WJ machinery fixed, the Student-t ($\nu = 5$) upgrade over the Normal baseline closed the in-sample excess-kurtosis gap from a 5.5 value under Normal emissions to a 7.6 value under Student-t emissions, against an observed 7.7 value, while leaving KS, AD, ACF-MAE, Wasserstein-1, and Hellinger essentially unchanged. The out-of-sample column shows the same ordering on tail moments and a narrower spread on distributional pass rates, consistent with the single-year ($T = 249$) sample.

F State-resolution sensitivity

We swept the partition resolution over $N \in \{30, 60, 90, 100, 150, 200, 350\}$ and re-evaluated the in-sample KS and AD pass rates for HMM-NJ and HMM-WJ. Pass rates were flat across $N \in \{60, \ldots, 200\}$ with sub-percent variation, indicating that the model was insensitive to partition resolution in this range. At $N = 30$ both models lost approximately 5 percentage points on the AD test, reflecting the coarser quantile bins' reduced ability to track tail behavior. At $N = 350$ several quantile bins were never visited in the historical record, leaving the corresponding emission parameters undefined; this set the practical upper bound on N for the SPY in-sample window. We adopted $N = 100$ as the operating point for all subsequent experiments.

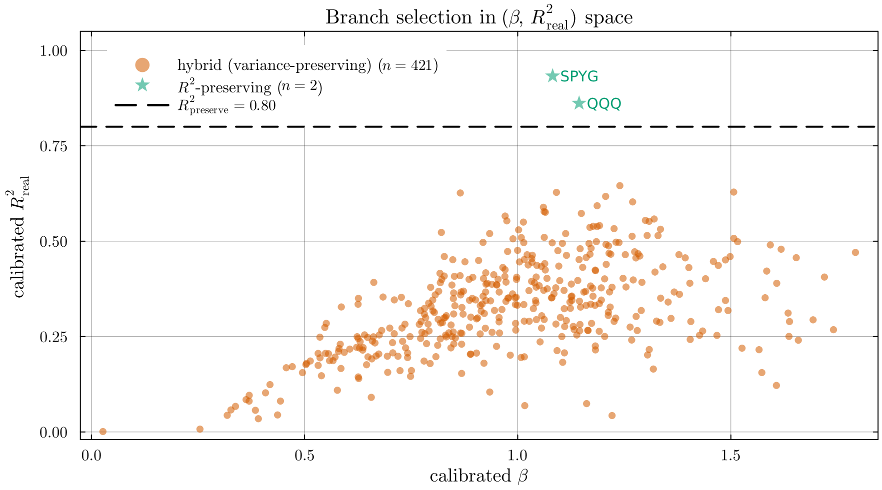

Figure S6: **Branch assignment in calibrated $(\beta, R^2_{\text{real}})$ space.** The $R^2_{\text{preserve}} = 0.80$ threshold (dashed line) separates tracker assets from single stocks. In this universe, only QQQ and SPYG exceed the threshold, and both are assigned to the R^2-preserving branch. The remaining 421 tickers populate the variance-preserving branch; the clipped variance-preserving branch is unused because the market loading ratio $\rho_i = \beta_i^2 \sigma_m^2 / \sigma_{\text{gen},i}^2$ stays below 0.90 for every ticker.

Table S5: **Aggregate metrics by β quartile, across composers.** Quartile cutoffs are computed from the calibrated β distribution across the 423-ticker universe. The hybrid KS pass rate declines gently from 67% in Q1 to 44% in Q4 as the market loading ratio ρ_i grows, consistent with the tail-attenuation caveat noted alongside Algorithm 2; naive collapses from 16% to under 1% over the same range.

β bucket	Composer	$\lvert\hat{\beta} - \beta\rvert$	KS pass (%)	W_1	κ	Var(g)
Q1 (low β)	Naive	0.018	15.7	0.414	8.971	18.1
	Gaussian SIM	0.015	0.0	0.620	0.622	14.0
	Hybrid	0.016	67.3	0.200	8.369	14.6
	JumpHMM-on-residuals	0.015	73.2	0.204	7.059	14.3
	Block bootstrap	0.015	71.3	0.208	6.286	13.9
	GARCH(1,1)-t	0.015	57.8	0.253	5.255	13.4
Q2	Naive	0.021	0.9	0.632	7.510	25.1
	Gaussian SIM	0.016	0.2	0.653	1.509	18.0
	Hybrid	0.017	45.9	0.254	6.889	19.0
	JumpHMM-on-residuals	0.017	82.7	0.200	7.799	18.7
	Block bootstrap	0.016	79.8	0.197	7.213	18.0
	GARCH(1,1)-t	0.015	76.4	0.227	6.470	17.8
Q3	Naive	0.023	0.4	0.802	7.390	33.3
	Gaussian SIM	0.017	1.4	0.688	1.947	23.4
	Hybrid	0.018	44.6	0.297	6.904	24.4
	JumpHMM-on-residuals	0.018	78.3	0.242	7.454	24.0
	Block bootstrap	0.017	75.0	0.236	6.888	22.8
	GARCH(1,1)-t	0.016	71.0	0.263	6.265	21.9
Q4 (high β)	Naive	0.029	0.8	0.976	7.233	51.2
	Gaussian SIM	0.022	0.6	0.825	2.054	36.3
	Hybrid	0.022	43.7	0.363	6.739	37.1
	JumpHMM-on-residuals	0.022	69.1	0.313	6.842	37.0
	Block bootstrap	0.021	67.2	0.311	6.334	35.3
	GARCH(1,1)-t	0.021	64.5	0.361	6.064	34.7

Table S6: **Clipping-branch stress test under market-volatility scaling.** The market growth-rate path was rescaled by $\gamma \in \{1, 2, 3\}$ with the calibrated marginals held fixed, and the full hybrid composer re-evaluated on the 423-ticker universe. Clipping is a stability mechanism: as γ rises the share of clipped tickers grows and the hybrid KS pass rate increases rather than degrading.

γ	**Clipped (%)**	**Hybrid KS (%)**
1	0.0	50.3
2	76.4	71.8
3	95.2	77.3

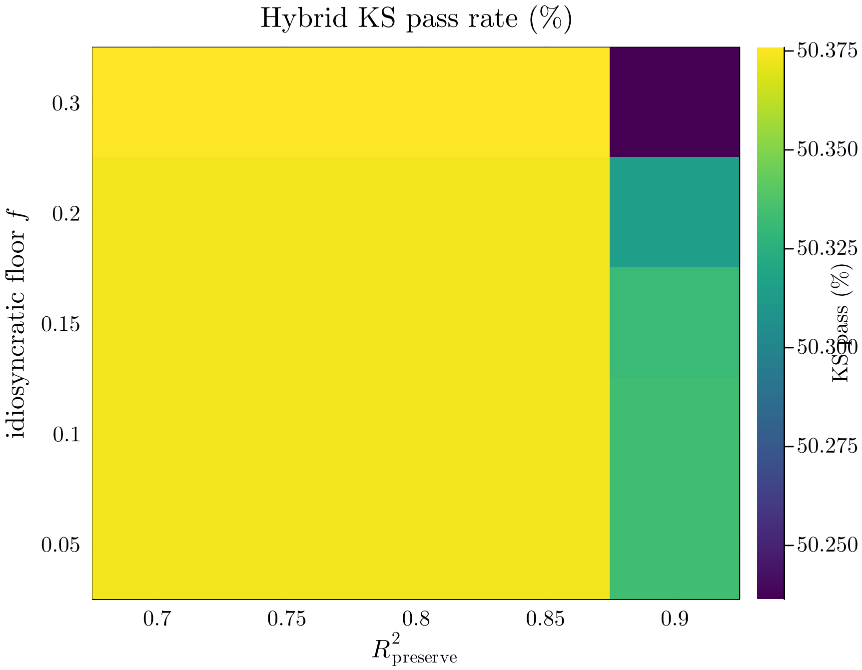

Figure S7: **Hybrid KS pass rate across a 5×5 hyperparameter grid.** Columns index the R^2-preserve threshold $R^2_{\text{preserve}} \in \{0.70, 0.75, 0.80, 0.85, 0.90\}$; rows index the idiosyncratic floor $f \in \{0.05, 0.10, 0.15, 0.20, 0.30\}$. Hybrid KS pass rate varies from 50.24% to 50.38% across the 25 cells, a spread of 0.14 percentage points. The operating point $(0.80, 0.10)$ used throughout the paper is indistinguishable from every other grid point within sampling noise, indicating that the headline result is not a data-peeked local optimum.

G Cross-asset generalization

To verify that the hybrid HMM framework generalized beyond SPY, we re-ran the full fit-and-validate pipeline on three single-stock tickers chosen to span sector and volatility regimes: NVDA (semiconductors, high volatility), JNJ (consumer staples, low volatility), and JPM (financials, moderate volatility). Per-ticker tune outputs at $(n_{\text{paths}} = 200, \text{seed} = 1234)$ are reported in Table S7. Two qualifications apply: the tune objective combines an ACF target and a kurtosis target through a fixed-weight sum, and the kurtosis penalty has substantial sample variance on heavy-tailed paths, so the specific argmin location is sensitive to Monte-Carlo noise at finite n_{paths}; the (ϵ^*, λ^*) values in the table are reproducible at the stated configuration but should not be over-interpreted as canonical per-ticker optima. The quantities the tune procedure produces reliably are the resulting distributional metrics at the chosen operating point: in-sample HMM-WJ KS pass rates exceeded 98.9% on all three tickers, indicating a close fit to each ticker's training-window distribution. Out-of-sample generalization on the 249-day 2025 hold-out varied by asset: NVDA retained 97.7% KS pass, comparable to its in-sample fit, while JNJ and JPM attenuated to 74.0% and 79.2% respectively, reflecting larger between-window shifts in the per-ticker marginal distribution than the SPY series exhibited.

Table S7: **Cross-asset generalization of HMM-WJ across three single-stock tickers spanning sector and volatility regimes.** Per-ticker tune outputs (ϵ^*, λ^*) at $n_{\text{paths}} = 200$ and seed $= 1234$, and the corresponding HMM-WJ pass rates on $1{,}000$ simulated paths at $\alpha = 0.05$ against each ticker's in-sample $(2014–2024)$ and out-of-sample (2025) growth-rate history. Standard errors in parentheses (binomial SE for pass rates). The (ϵ^*, λ^*) values are reproducible at the stated tune configuration but the composite objective has multiple near-equivalent local minima, so they are reported as the operating points used for the IS / OoS fit metrics rather than as canonical per-ticker optima.

Ticker	ϵ^*	λ^*	**IS KS (%)**	**IS AD (%)**	**OoS KS (%)**	**OoS AD (%)**
NVDA (semiconductors)	10^{-4}	70	98.9 (0.3)	96.7 (0.6)	97.7 (0.5)	96.1 (0.6)
JNJ (consumer staples)	10^{-4}	70	99.4 (0.2)	96.3 (0.6)	74.0 (1.4)	61.3 (1.5)
JPM (financials)	10^{-4}	80	99.1 (0.3)	95.7 (0.6)	79.2 (1.3)	77.7 (1.3)

Table S8: **Seed-uncertainty summary across** 8 **random seeds.** Each cell reports mean ± SD across the 8 seeds in $\{1234, 2345, 3456, 4567, 5678, 6789, 7890, 8910\}$, with each seed running the full 423-ticker × 100-replication protocol. Conditional-variance paths for GARCH(1,1)-t are re-simulated at scoring time so its seed SD is informative. The between-composer differences in KS pass rate exceed the seed-induced SD by roughly two orders of magnitude (hybrid SD 0.18 pp vs 25 pp gap to JumpHMM-on-residuals).

Composer	$\lvert\hat{\beta} - \beta\rvert$	$\lvert\hat{R}^2 - R^2_{\text{real}}\rvert$	KS pass (%)	AD pass (%)	κ	Hill
Naive	0.022 ± 0.000	0.088 ± 0.000	4.41 ± 0.07	2.71 ± 0.02	7.746 ± 0.015	0.369 ± 0.000
Gaussian SIM	0.017 ± 0.000	0.008 ± 0.000	0.60 ± 0.04	0.50 ± 0.01	1.428 ± 0.005	0.217 ± 0.000
Hybrid	0.018 ± 0.000	0.021 ± 0.000	50.40 ± 0.18	47.50 ± 0.19	7.150 ± 0.011	0.364 ± 0.000
JumpHMM-on-residuals	0.018 ± 0.000	0.015 ± 0.000	75.61 ± 0.11	70.51 ± 0.12	7.250 ± 0.013	0.365 ± 0.000
Block bootstrap	0.017 ± 0.000	0.020 ± 0.000	73.24 ± 0.09	66.79 ± 0.11	6.676 ± 0.011	0.330 ± 0.000
GARCH(1,1)-t	0.017 ± 0.000	0.036 ± 0.000	66.96 ± 0.30	61.00 ± 0.26	6.072 ± 0.022	0.318 ± 0.000

H SIM composition derivations and properties verification

The variance-corrected single-index composition states two closed-form results, the variance scale of Eq. (7) and the residual-variance target of Eq. (10), and asserts that the composed series g_i satisfies the three preservation criteria (SIM recovery, marginal fidelity, cross-sectional dependence) in the large-T limit. To derive Eq. (7), form the candidate residual $\varepsilon_i = s_i\, \tilde{g}_i$ for some scalar $s_i \in (0, 1]$ and require that the composed series have the same variance as the generator path:

$$\mathrm{Var}(g_i) \;=\; \mathrm{Var}(\alpha_i + \beta_i\, g_m + \varepsilon_i) \;=\; \mathrm{Var}(\beta_i\, g_m + \varepsilon_i) \;=\; \sigma^2_{\text{gen},i}, \tag{S18}$$

where α_i drops out because it is a constant offset. Expanding the variance of the sum:

$$\mathrm{Var}(\beta_i\, g_m + \varepsilon_i) \;=\; \beta_i^2\; \mathrm{Var}(g_m) \;+\; 2\beta_i\; \mathrm{Cov}(g_m,\, \varepsilon_i) \;+\; \mathrm{Var}(\varepsilon_i). \tag{S19}$$

Substituting $\varepsilon_i = s_i\, \tilde{g}_i$ yields $\mathrm{Var}(\varepsilon_i) = s_i^2\, \sigma^2_{\text{gen},i}$ and $\mathrm{Cov}(g_m,\, \varepsilon_i) = s_i\; \mathrm{Cov}(g_m,\, \tilde{g}_i)$, so under the independence assumption $\mathrm{Cov}(\tilde{g}_i,\, g_m) \approx 0$ the cross term vanishes and the variance condition reduces to $\beta_i^2\, \sigma_m^2 + s_i^2\, \sigma^2_{\text{gen},i} = \sigma^2_{\text{gen},i}$, which rearranges directly to Eq. (7).

To derive Eq. (10), write the population R^2 identity for Eq. (4):

$$R^2 \;=\; \frac{\beta_i^2\, \sigma_m^2}{\beta_i^2\, \sigma_m^2 + \sigma^2_{\varepsilon,i}}, \tag{S20}$$

and invert for the residual variance that hits the calibrated R^2; the result is the boxed Eq. (10). By construction the composed series g_i then satisfies the three preservation criteria in the large-T limit. OLS of g_i on g_m recovers $\hat{\alpha}_i \to \alpha_i$ and $\hat{\beta}_i \to \beta_i^{\text{eff}}$, with $\beta_i^{\text{eff}} = \beta_i$ unclipped and β_i^{eff} persisted alongside β_i when the clip triggers. Marginal variance matches $\sigma^2_{\text{gen},i}$ on the variance-preserving branch, either exactly $\mathrm{Var}(g_i) = \beta_i^2 \sigma_m^2 + s_i^2 \sigma^2_{\text{gen},i} = \sigma^2_{\text{gen},i}$, or by the choice of β_i^{eff} in Eq. (9) under clipping; on the R^2-preserving branch the marginal variance instead matches $\beta_i^2 \sigma_m^2 / R^2_{i,\text{real}}$, the value implied by the calibrated $(\beta_i, R^2_{i,\text{real}})$ rather than by the generator marginal. Tail behavior on the variance-preserving branch follows from the floor: because at least $f\, \sigma^2_{\text{gen},i}$ of the variance is reserved for ε_i, the tails inherited from $\tilde{g}_i$ dominate whenever $(\beta_i^{\text{eff}})^2\sigma_m^2$ is small relative to those tails, while as β_i grows the marginal of g_i tightens toward Gaussian along the market-driven axis, mildly for low-β assets and noticeably at high β.